

Knots, black holes, databases, and birthdays: Collision entropy of knot invariants

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A knot invariant can assign the same value to inequivalent knots. We prove that, under two rooted random-diagram models, the normalized Alexander polynomial distinguishes two independent diagrams with n crossings with probability $1 - O(n^{-1/2})$, and every fixed polynomial value has exponentially small probability. With exponentially high probability, the diagram shadow contains linearly many disjoint trefoil patterns. Conditional on the shadow and exterior crossing choices, these patterns contribute a binomial term to the exponent of 3 in the determinant. For a discrete invariant I , let $\alpha_n(I)$ denote the probability that two independent samples have the same value. Its negative logarithm is the collision entropy. We prove that $\alpha_n(I)^{-1/2}$ is the birthday scale for independent databases: repetitions are unlikely far below this scale and occur with probability tending to one far above it. We also derive exact formulas for sampling finite collections without replacement and for the expected cost of applying successive invariants. These formulas apply to other databases whose records are compared through discrete summaries. In knot comparison, $\alpha_n(I)$ is the expected fraction of pairs left unresolved by the invariant. On a balanced pair-classification benchmark, testing equality of Alexander polynomials gives accuracy tending to one. The errors are exactly the inequivalent pairs with equal polynomial values.

knot invariants | collision entropy | birthday paradox | random knot diagrams | computational black holes

A knot is a closed curve embedded in three-dimensional space. A knot diagram records a projection of that curve to the plane, together with the over-under information at each crossing. Two diagrams represent the same knot when the corresponding curves can be deformed into one another in three dimensions without cutting a strand or passing one strand through another. The basic comparison problem is to decide whether two given diagrams represent the same knot.

Significance Statement

The Alexander polynomial is a computable quantity used to distinguish knots, although different knots can share its value. Under two models of random knot diagrams, we prove that independent samples have different Alexander polynomials with probability approaching one as the number of crossings grows. In sufficiently large collections, repetitions become likely. We relate the size at which repetitions appear to the probability of a match between two samples, as in the birthday problem. The resulting formulas apply to many other databases whose records are compared through discrete summaries, including hashes. They determine the expected proportion of comparisons requiring further tests and explain how high classification accuracy can coexist with persistent failures on particular pairs.

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A knot invariant assigns a value to the represented knot. Equivalent knots must have the same value, so different values certify inequivalence. Equal values need not certify equivalence: an incomplete invariant assigns the same value to some distinct knots. The normalized Alexander polynomial is a classical invariant of this kind. Throughout, we use the symmetric normalization $\Delta_K(t) = \Delta_K(t^{-1})$ and $\Delta_K(1) = 1$. We ask whether an incomplete invariant can separate almost every random pair and still repeat somewhere in a sufficiently large database.

We work with two probability laws on diagrams having exactly n crossings. In each model a diagram is rooted by marking one directed half-edge of the underlying planar map. Under either law, the Alexander polynomial separates two independent diagrams with probability $1 - O(n^{-1/2})$, and the probability of every fixed normalized Alexander polynomial decays exponentially. These assertions concern the specified sampling laws and do not imply injectivity. A database of M independently sampled diagrams contains $\binom{M}{2}$ pairs. Since this number grows quadratically, a repeated value can occur with probability tending to one even while a single random pair is separated with probability tending to one. Pairwise separation and database uniqueness therefore have different thresholds.

The same distinction applies to any discrete fingerprint. Fix n , let D_n be a random object of complexity n , and let $I(D_n)$ take values in a finite or countable set. Here I is an invariant: equivalent objects receive the same value. A fingerprint is either one such map I or a finite tuple $J = (I_1, \dots, I_k)$ of them. Its level sets are the fingerprint fibers. We write

$$p_v = \mathbb{P}\{I(D_n) = v\}, \quad \alpha_n(I) = \sum_v p_v^2. \quad [1]$$

For an independent draw D'_n from the same law, the collision probability is,

$$\alpha_n(I) = \mathbb{P}\{I(D_n) = I(D'_n)\}.$$

Its negative (natural) logarithm,

$$H_2(I(D_n)) = -\log \alpha_n(I),$$

is the order-two Rényi entropy, also called the collision entropy (1). When unequal invariant values prove non-equivalence, $1 - \alpha_n(I)$ is the probability that the invariant resolves one random comparison immediately.

The equality event also has a computational meaning. An invariant-first algorithm computes I , returns NO when the two values differ, and invokes a complete equivalence procedure when they agree. The second step is necessary because the invariant alone does not decide pairs in a common fiber. We call this equality region the *invariant black hole*. Generic-case complexity studies algorithms that are efficient on an asymptotically dominant set while exceptional inputs retain the unresolved behavior (2–5); later work calls such an exceptional set the “black hole of the algorithm” (6). For independent inputs, its probability is $\alpha_n(I)$. The term is algorithm-relative: membership says nothing about intrinsic difficulty.

Now sample an independent database $D_n^{(1)}, \dots, D_n^{(M)}$ of copies of D_n and let

$$C_M(I) = \sum_{1 \leq i < j \leq M} \mathbf{1}_{\{I(D_n^{(i)}) = I(D_n^{(j)})\}}$$

count colliding pairs. Then

$$\mathbb{E}C_M(I) = \binom{M}{2} \alpha_n(I). \quad [2]$$

The random variable $C_M(I)$ is also the number of complete-procedure calls made by the invariant-first algorithm in an all-pairs comparison. Thus $\alpha_n(I)$ is the random-pair equality probability, the probability of entering the invariant black hole, and the factor in the expected database workload $\binom{M}{2} \alpha_n(I)$.

Since an M -record database has $\binom{M}{2}$ pairs, the preceding expectation suggests a threshold when $M^2 \alpha_n(I)$ is of constant order. The universal law proved below makes this precise. Its distribution-free birthday scale is

$$M_B(n) = \alpha_n(I)^{-1/2} = \exp\left(\frac{H_2(I(D_n))}{2}\right). \quad [3]$$

Databases of size $o(M_B(n))$ are collision-free with high probability, whereas databases of size $\omega(M_B(n))$ contain a collision with high probability. Thus, if $\alpha_n(I) \rightarrow 0$ and $M_n^2 \alpha_n(I) \rightarrow \infty$, a random pair is separated with probability tending to one while a database of size M_n contains a repeated fingerprint with probability tending to one. This is the birthday phenomenon for an arbitrary discrete law, expressed in terms of collision entropy (7, 8).

We now return to knots. For a knot K , its determinant is $\det K = |\Delta_K(-1)|$. Equality of normalized Alexander polynomials therefore implies equality of determinants. If the Alexander polynomial equals a fixed polynomial P , then the determinant equals the fixed integer $|P(-1)|$. Thus a uniform bound on determinant probabilities controls both equality between two random Alexander polynomials and equality with any fixed Alexander polynomial.

We prove this determinant bound in two rooted random-diagram models (9, 10). The connected sum of two knots is formed by removing a short arc from each and joining the resulting pairs of endpoints. Its determinant is the product of their determinants; a trefoil has determinant 3, and the unknot has determinant 1. The shadow of a diagram is its knot projection with the crossing information removed. An opened-trefoil slot is a small region whose internal crossing signs can produce either a trefoil summand or an unknot summand. With exponentially high probability, the shadow contains linearly many disjoint such slots. After the shadow and all exterior crossing signs have been exposed, the signs inside the slots remain independent. Cutting along a sphere that meets the knot in two points separates a local knot factor. The selected slots admit a single family of pairwise disjoint

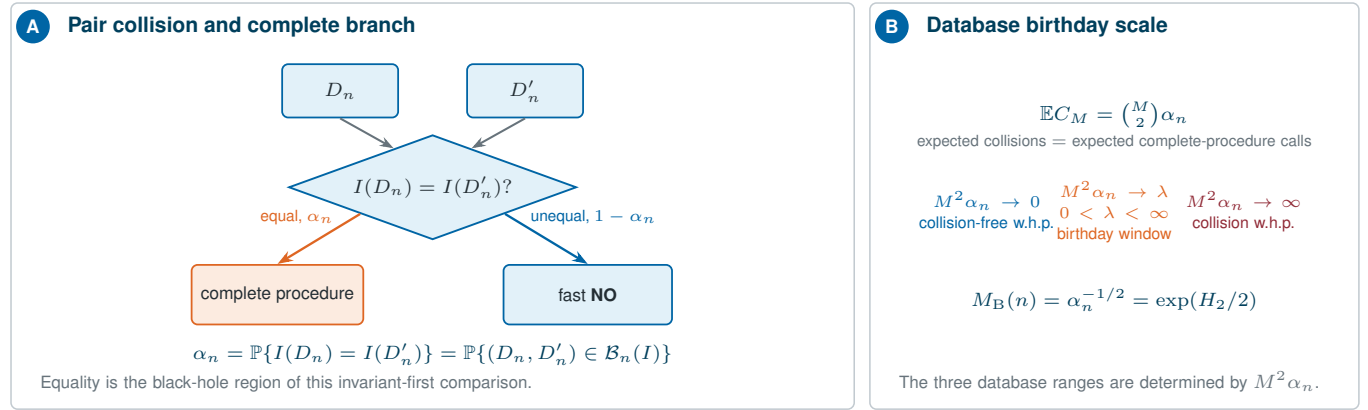


Fig. 1. Collision probability for a pair and a database. (A) The invariant I assigns a fingerprint value to each object. For two independent objects, $\alpha_n = \mathbb{P}\{I(D_n) = I(D'_n)\}$ is the probability that their values agree. Unequal values certify inequivalence and give a fast NO. Equal values leave the comparison unresolved by I , so an invariant-first algorithm calls its complete procedure. The term black hole refers only to this algorithm-relative equality region; it does not imply intrinsic hardness. (B) For M independent records, the expected number of colliding pairs, and hence of complete-procedure calls in an all-pairs comparison, is $\mathbb{E}C_M = \binom{M}{2} \alpha_n$. The database is collision-free with high probability when $M^2 \alpha_n \rightarrow 0$ and contains a collision with high probability when $M^2 \alpha_n \rightarrow \infty$. The critical window is $M^2 \alpha_n \rightarrow \lambda \in (0, \infty)$. The birthday scale is $M_B(n) = \alpha_n^{-1/2} = \exp(H_2/2)$, where $H_2 = -\log \alpha_n$ is the collision entropy.

such spheres for every internal redecoration, and replacing the selected tangles by fixed trivial arcs gives the same exterior knot for every internal sign vector. Each slot contributes a trefoil with conditional probability $1/4$ and an unknot with conditional probability $3/4$. The number of trefoil outcomes is a binomial coordinate in the 3-adic valuation, the exponent of 3 in the determinant. Consequently,

$$\sup_{a \geq 1} \mathbb{P}\{\det K(D_n) = a\} = O(n^{-1/2}). \quad [4]$$

It follows from Eq. (4) that two independent random diagrams have equal determinants with probability $O(n^{-1/2})$, and a determinant-first comparison invokes its complete procedure with vanishing probability. For a fixed target, determinant equality forces the binomial coordinate into a fixed lower tail, so this probability decays exponentially. For independent databases, the same estimate gives

$$M_n = o(n^{1/4}) \implies \mathbb{P}\{\text{any determinant-first fallback call}\} \rightarrow 0.$$

This is a sufficient condition obtained from the upper bound $\alpha_n(\det) = O(n^{-1/2})$; no converse at the exponent $1/4$ is proved. The birthday scale is $M_B(n) = \alpha_n(\det)^{-1/2}$. The bound gives $M_B(n) = \Omega(n^{1/4})$, but its order remains unknown.

The preceding database model uses independent samples from a probability law. A finite knot census is different: it is a fixed population of distinct entries sampled without replacement. If the census has T entries and invariant-fiber sizes $m_1, \dots, m_R$, the no-collision probability for a uniform M -subset is $e_M(m_1, \dots, m_R) / \binom{T}{M}$, where e_M is the M th elementary symmetric polynomial. Its numerator counts collision-free subsets by choosing one entry from each of M distinct fibers. The number of occupied values alone is insufficient: a few large fibers can cause early collisions even when many values occur. An invariant cascade applies a sequence of invariants, sending only records that agree on all earlier values to the next test. Fiber sizes determine the expected numbers of unresolved objects and pairs after each stage, yielding a two-stage ordering criterion and a marginal break-even criterion.

Recent work studies related finite-population statistics. Dłotko, Gurnari, and Sazdanović introduce Mapper-type methods in (11); their subsequent knot-theoretic study attributes to Ernesto Lupercio the observation that unique-value ratios may decrease even while the probability of separating two random knots tends to one (12). Tubbenhauer and Zhang compare quantum invariants on cumulative prime-knot censuses using value counts and repeated-pair experiments (13). Kelomäki et al. prove exponential decay of value and singleton-detection ratios, together with exponential growth of typical fibers, in cumulative prime-alternating-link populations (14). Our knot-specific theorem instead concerns two-sample collision probabilities in rooted fixed-crossing ensembles and proves a moving-target determinant bound.

The argument uses established results on coincidence probabilities, Rényi entropy, generic-case complexity, knot invariants, and random-diagram patterns (1–6, 10). Its new input is the conditional binomial coordinate in the determinant valuation. This gives the anti-concentration estimate and its consequences for random pairs, databases, and invariant-first computation.

1. Collision entropy and the inverse birthday law

The pair-versus-database transition depends only on the discrete distribution of the fingerprint, not on knot theory. We therefore prove the birthday law for an arbitrary discrete fingerprint distribution.

For each n , let $X_n = I(D_n)$ have a discrete law, and fix this ambient n throughout the following definitions. Write X'_n for an independent copy and $X_n^{(1)}, \dots, X_n^{(M)}$ for independent copies of X_n , and set

$$p_v = \mathbb{P}\{X_n = v\}, \quad \alpha_n = \sum_v p_v^2, \quad \beta_n = \sum_v p_v^3, \quad p_n^{\max} = \sup_v p_v, \quad C_M = \sum_{i < j} \mathbf{1}_{\{X_n^{(i)} = X_n^{(j)}\}}.$$

Thus α_n is the probability that two independent samples agree, and β_n is the probability that three independent samples agree. The largest atom satisfies

$$(p_n^{\max})^2 \leq \alpha_n \leq p_n^{\max},$$

so $\alpha_n \rightarrow 0$ if and only if $p_n^{\max} \rightarrow 0$. Counting covariances gives

$$\begin{aligned}\mathbb{E}C_M &= \binom{M}{2} \alpha_n, \\ \text{Var}(C_M) &= \binom{M}{2} (\alpha_n - \alpha_n^2) + 6 \binom{M}{3} (\beta_n - \alpha_n^2).\end{aligned}$$

To prove that a collision occurs with high probability, one needs concentration of the collision count. The only pair comparisons that can be dependent are those sharing one sampled object, and their joint probability is β_n ; comparisons on disjoint pairs are independent. Thus $M^2\alpha_n$, the expected collision count up to a constant factor, is the parameter governing the transition. The moment calculation and the scale proof are in SI Appendix, Theorem S2.2 and Theorem S2.3. The next theorem shows that this parameter determines the answer away from the transition scale.

Theorem 1 (Universal inverse birthday law). *For each n , let the fingerprint law be any discrete distribution, possibly depending on n . For any integer sequence $M_n \geq 2$, let C_{M_n} denote the collision count among M_n iid samples from the law at level n . Then*

$$M_n^2 \alpha_n \rightarrow 0 \implies \mathbb{P}\{C_{M_n} > 0\} \rightarrow 0,$$

whereas

$$M_n^2 \alpha_n \rightarrow \infty \implies \mathbb{P}\{C_{M_n} > 0\} \rightarrow 1.$$

Equivalently, with

$$M_B(n) := \alpha_n^{-1/2} = \exp\left(\frac{H_2(X_n)}{2}\right),$$

databases of size $o(M_B(n))$ are collision-free with high probability, while databases of size $\omega(M_B(n))$ contain a collision with high probability.

Proof. The first implication follows from Markov's inequality and $\mathbb{E}C_{M_n} = \binom{M_n}{2} \alpha_n$. For the second, the power-sum inequalities

$$\alpha_n^2 \leq \beta_n \leq \alpha_n^{3/2}$$

imply

$$\frac{\text{Var}(C_M)}{(\mathbb{E}C_M)^2} \leq \frac{4}{M^2 \alpha_n} + \frac{4}{M \sqrt{\alpha_n}}.$$

If $M_n^2 \alpha_n \rightarrow \infty$, both terms vanish. Chebyshev's inequality then gives $\mathbb{P}\{C_{M_n} = 0\} \rightarrow 0$. $\square$

The threshold occurs when M_n is comparable to $\alpha_n^{-1/2}$. At that scale the expected collision count remains bounded, so the preceding zero-one law does not determine the limiting probability. If $M_n^2 \alpha_n \rightarrow \lambda \in (0, \infty)$ and, in addition,

$$M_n^3 \beta_n \rightarrow 0,$$

then $M_n \rightarrow \infty$, $\mathbb{E}C_{M_n} \rightarrow \lambda/2$, and

$$C_{M_n} \implies \text{Poisson}(\lambda/2).$$

The proof is in SI Appendix, Theorem S2.5. The construction in SI Appendix, Proposition S2.6 has a moving heavy atom and fails to have this Poisson limit, so the additional condition cannot be omitted in general. Such failures are familiar in Poisson approximation with local dependence or rare clusters (15, 16).

Corollary 2 (Inverse birthday phenomenon). *Suppose $\alpha_n \rightarrow 0$ and $M_n^2 \alpha_n \rightarrow \infty$. Then*

$$\mathbb{P}\{X_n \neq X'_n\} \rightarrow 1, \quad \mathbb{P}\{C_{M_n} > 0\} \rightarrow 1.$$

Thus one random pair is asymptotically separated while a database of size M_n asymptotically contains a repeated fingerprint.

For knot invariants, it remains to bound $p_n^{\max}$ uniformly. The number of occupied invariant values alone gives no such bound, since a few fibers may account for most of α_n . A bound for each fixed value would also be insufficient: the most likely value may change with n . The following section controls the supremum over all determinant values, including values that move with n , using trefoil-slot contributions that are independent after conditioning on the shadow and the exterior crossing signs.

2. Trefoil slots and determinant anti-concentration

A knot invariant has the same value on every diagram of a given knot, although different knots may share that value. For a random diagram, anti-concentration means that no single value of the invariant has substantial probability. We prove such an estimate for the knot determinant,

$$\det K := |\Delta_K(-1)|,$$

the absolute value of the Alexander polynomial evaluated at -1 , in two random-diagram ensembles introduced by Chapman (10). For a positive integer m , the 3-adic valuation $v_3(m)$ is the exponent of 3 in its prime factorization. With exponentially high probability, the unsigned planar diagram contains linearly many separated places where three unexposed crossing choices can insert either a trefoil or an unknot. These choices produce a binomial summand in $v_3(\det K)$ and hence prevent the determinant from concentrating at one integer.

2.1. Random signs on a fixed shadow. Let $\mathcal{D}_n^{\text{gen}}$ be Chapman's rooted general one-component diagrams with n crossings and let $\mathcal{D}_n^{\text{red}}$ be the corresponding rooted reduced class. A root is a distinguished dart of the planar map, and this mark is retained in the counting; one-component means that the diagram represents a knot rather than a link. Write $\mathcal{S}_n^*$ for the shadow class obtained from $\mathcal{D}_n^*$ by forgetting which strand passes over at each crossing, where $\star \in \{\text{gen}, \text{red}\}$. Thus a shadow is a rooted four-valent planar map with the strand continuation retained. The reduced class excludes shadows with a disconnecting crossing vertex. The one-component and reduced conditions are properties of the shadow. Restoring one of the two over-under choices at each vertex gives a decorated lift. Every rooted shadow in either class therefore has exactly 2^n decorated lifts, and a uniform diagram D_n can be sampled exactly by

$$\text{uniform rooted shadow} + \text{independent fair signs at its } n \text{ crossings.}$$

The slots are selected from the shadow before any internal signs are exposed. The product law is proved in SI Appendix, Lemma S1.1.

Start with the standard three-crossing trefoil diagram and cut one edge away from its crossings. The result is a one-strand tangle with two boundary ends; Q denotes its shadow. Rejoining the ends gives the doubled-triangle graph. The tangle is prime, two-leg-prime, and reduced and admits an asymmetric fixed decoration P , as required by Chapman's theorem.

Chapman's connect-sum construction and pattern theorem imply that P occurs linearly many times as a connected-sum tangle, outside an exponentially small exceptional set (10, Propositions 3 and 10). Forgetting the signs of P leaves the same abundance of unsigned Q -slots. A deterministic rule based only on the rooted planar map then selects a separated subfamily: after occurrences within bounded graph distance are removed, it retains

$$s_n = \lfloor \delta n \rfloor$$

Q -slots whose three-vertex carriers are at pairwise graph distance at least three, on an event $\mathcal{G}_n$ satisfying

$$\mathbb{P}(\mathcal{G}_n^c) \leq C_0 e^{-c_0 n}.$$

Apply Chapman's theorem to P , then forget its crossing signs. Admissibility, passage to shadows, and shadow-measurable selection are proved in SI Appendix, Lemma S4.1, Lemma S4.2, and Lemma S4.3.

2.2. The Bernoulli contribution of one slot. Work on $\mathcal{G}_n$, where the selected family has exactly s_n slots. Reveal the complete shadow and every crossing sign outside their three-vertex carriers, but leave the three signs in each slot unexposed. Let $\mathcal{F}_n$ be the sigma-field generated by the revealed data. The event $\mathcal{G}_n$, the selected list, and the separator geometry are $\mathcal{F}_n$ -measurable. Conditional on $\mathcal{F}_n$, the $3s_n$ internal signs remain independent and fair.

For the j th slot, write the reference-relative signs as $\varepsilon_{n,j} = (\varepsilon_{n,j,1}, \varepsilon_{n,j,2}, \varepsilon_{n,j,3})$. There are eight equally likely triples. The two unanimous triples close to the two mirror images of the trefoil; each of the other six closes to the unknot. Let $T(2, r)$ denote the closure of the two-strand braid of exponent r . After a two-braid generator is chosen consistently with the embedding, the local closure is

$$T(2, \eta_j(\varepsilon_{n,j,1} + \varepsilon_{n,j,2} + \varepsilon_{n,j,3})), \quad \eta_j \in \{-1, +1\},$$

where η_j is fixed by the shadow. Hence

$$Y_{n,j} := \mathbf{1}\{\varepsilon_{n,j,1} = \varepsilon_{n,j,2} = \varepsilon_{n,j,3}\}, \quad \mathcal{L}((Y_{n,1}, \dots, Y_{n,s_n}) \mid \mathcal{F}_n) = \text{Bernoulli}(1/4)^{\otimes s_n}$$

almost surely on $\mathcal{G}_n$.

This local calculation does not by itself give a simultaneous connected-sum decomposition. For that, the cutting spheres must be pairwise disjoint and must remain fixed when any collection of internal signs is changed. Each selected occurrence contains its three crossing vertices and all five edges between them; exactly two distinct joining edges connect it to the rest of the shadow. Distinct internal carriers lie at graph distance at least three. These facts produce a single shadow-measurable family of pairwise disjoint balls that separates all selected occurrences for every choice of their internal crossing signs.

With the shadow and exterior signs fixed, let $D_n[\mathbf{e}]$ denote the diagram obtained by assigning an arbitrary internal sign vector $\mathbf{e} = (\mathbf{e}_1, \dots, \mathbf{e}_{s_n})$. Replace the strand inside each separating ball by a fixed trivial arc. The resulting knot is the exterior knot K_n^{ext} ; it is $\mathcal{F}_n$ -measurable and unchanged by $\mathbf{e}$. Cutting along the separating spheres and capping both sides expresses the original knot as a connected sum, denoted by $\#$. Simultaneously for all 2^{3s_n} choices,

$$K(D_n[\mathbf{e}]) \cong K_n^{\text{ext}} \# L_{n,1}(\mathbf{e}_1) \# \dots \# L_{n,s_n}(\mathbf{e}_{s_n}), \quad [5]$$

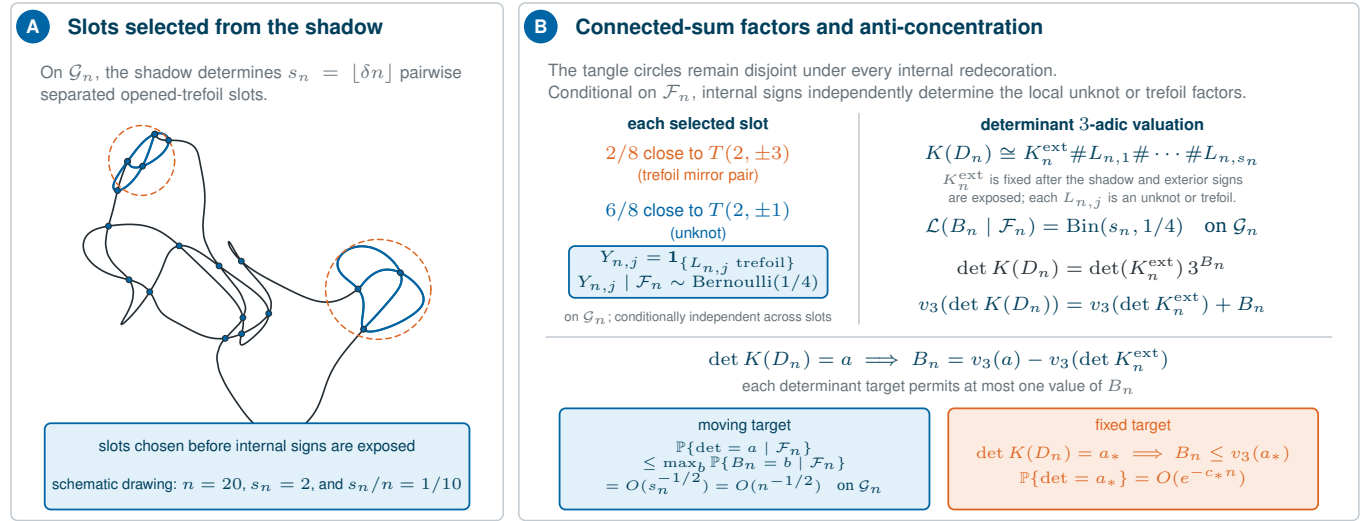


Fig. 2. Opened-trefoil slots and the binomial coordinate. A shadow is the four-valent planar map obtained from a knot diagram by forgetting the over-under choice at each crossing. (A) On the event $\mathcal{G}_n$, a rule depending only on the shadow selects $s_n = \lfloor \delta n \rfloor$ pairwise separated three-crossing slots before their signs are exposed. The schematic has $n = 20$ and $s_n = 2$, so its displayed ratio $1/10$ does not specify the theorem's constant $\delta > 0$. (B) The displayed tangle circles correspond to pairwise disjoint separating spheres that work for every later choice of the internal signs. After the shadow and all exterior signs have been revealed, $\mathcal{F}_n$ records this information. Two of the eight equally likely sign triples close to the trefoil mirror pair and six close to the unknot. The slot indicators are therefore conditionally independent Bernoulli($1/4$) variables, and their sum B_n , the number of trefoil factors, satisfies $\mathcal{L}(B_n | \mathcal{F}_n) = \text{Bin}(s_n, 1/4)$. Replacing every local tangle by a trivial arc gives the exterior knot K_n^{ext} . Cutting along the separating spheres gives $K(D_n) \cong K_n^{\text{ext}} \# L_{n,1} \# \cdots \# L_{n,s_n}$, where $\#$ denotes connected sum. The determinant $\det K = |\Delta_K(-1)|$ is multiplicative under connected sum; it equals 3 for a trefoil and 1 for the unknot. Hence $v_3(\det K(D_n)) = v_3(\det K_n^{\text{ext}}) + B_n$, where $v_3(m)$ is the exponent of 3 in m . Equality with a determinant target a forces $B_n = v_3(a) - v_3(\det K_n^{\text{ext}})$, whose largest conditional atom is $O(n^{-1/2})$. For fixed a_* , determinant equality implies $B_n \leq v_3(a_*)$, and this lower-tail probability is exponentially small.

where each local factor is an unknot or a trefoil according to the preceding rule. The two-edge boundary, distance-three separation, simultaneous separator balls, two-braid closure, and factorization for every internal sign vector are proved in SI Appendix, Lemma S4.5, Corollary S4.6, Lemma S4.7, Lemma S4.8, and Proposition S4.9.

The determinant is multiplicative under connected sum. The unknot has determinant 1, while either trefoil has determinant 3. For the random internal vector, put $B_n = \sum_{j=1}^{s_n} Y_{n,j}$, the number of trefoil factors. It follows, almost surely on $\mathcal{G}_n$, that

$$\det K(D_n) = \det(K_n^{\text{ext}}) 3^{B_n}, \quad v_3(\det K(D_n)) = v_3(\det K_n^{\text{ext}}) + B_n, \quad [6]$$

with

$$\mathcal{L}(B_n | \mathcal{F}_n) = \text{Bin}(s_n, 1/4).$$

The same K_n^{ext} works for every internal sign vector. Conditional on $\mathcal{F}_n$, $v_3(\det K_n^{\text{ext}})$ is fixed, and the only varying term in the valuation identity is B_n , with the displayed binomial law.

Theorem 3 (Determinant anti-concentration). *There are constants $\delta, c_0, C_0 > 0$ and $n_0 \geq 4$ such that, for $n \geq n_0$ and a uniform rooted diagram in either diagram model,*

$$\sup_{a \geq 1} \mathbb{P}\{\det K(D_n) = a\} \leq C_0 e^{-c_0 n} + \sqrt{\frac{2\pi}{3\lfloor \delta n \rfloor}} = O(n^{-1/2}).$$

The estimate is uniform over determinant targets $a = a_n$ that may vary with n .

Proof. Fix a target a . On $\mathcal{G}_n$, condition on $\mathcal{F}_n$. Eq. (6) forces B_n to equal the single $\mathcal{F}_n$ -measurable integer

$$v_3(a) - v_3(\det K_n^{\text{ext}}),$$

when that integer is admissible. If $B \sim \text{Bin}(m, 1/4)$, Fourier inversion gives

$$\sup_k \mathbb{P}\{B = k\} \leq \sqrt{\frac{2\pi}{3m}}.$$

Thus the conditional target probability is bounded by this quantity on $\mathcal{G}_n$. Taking expectations and adding $\mathbb{P}(\mathcal{G}_n^c)$ proves the claim. $\square$

If $p_a = \mathbb{P}\{\det K(D_n) = a\}$, then

$$\alpha_n(\det) = \sum_a p_a^2 \leq \sup_a p_a = O(n^{-1/2}),$$

and hence

$$H_2(\det K(D_n)) \geq \frac{1}{2} \log n - O(1).$$

In particular, two independent diagrams have different determinants with probability $1 - O(n^{-1/2})$. Section 3 applies this estimate to black-hole probabilities and databases of independent diagrams.

3. Vanishing computational black holes

The determinant bound has an algorithmic interpretation. Let I be a knot invariant, so $I(D) \neq I(D')$ certifies inequivalence, and let **Decide** be a complete knot-equivalence procedure. They give the invariant-first algorithm

$$A_I(D, D') = \begin{cases} \text{No}, & I(D) \neq I(D'), \\ \text{Decide}(D, D'), & I(D) = I(D'). \end{cases} \quad [7]$$

For each complexity n , let Ω_n denote the space of diagram inputs and let μ_n be the probability law from which a diagram is drawn. Define the *invariant black hole*

$$\mathcal{B}_n(I) := \{(D, D') \in \Omega_n^2 : I(D) = I(D')\}. \quad [8]$$

Generic-case complexity was introduced for decision problems by Kapovich et al. and developed in subsequent work on decidable and undecidable problems, including the halting problem (2–5). Later work uses the phrase “black hole of the algorithm” for the exceptional input set (6). Here $\mathcal{B}_n(I)$ means only the set of pairs on which A_I calls **Decide**. It may contain equivalent knots, inequivalent knots, easy instances, and difficult instances; membership says nothing about intrinsic hardness.

Proposition 4 (Black-hole and workload identities). *Let D_n, D'_n be independent with law μ_n . Then*

$$\mathbb{P}\{(D_n, D'_n) \in \mathcal{B}_n(I)\} = \mathbb{P}\{I(D_n) = I(D'_n)\} = \alpha_n(I). \quad [9]$$

For an independent database $D_n^{(1)}, \dots, D_n^{(M)}$ of copies of D_n , let N_v be the observed number of objects with invariant value v . The number $F_M(I)$ of all-pairs fallback calls made by Eq. (7) is exactly

$$F_M(I) = \sum_v \binom{N_v}{2} = C_M(I), \quad \mathbb{E}F_M(I) = \binom{M}{2} \alpha_n(I). \quad [10]$$

Proof. The event in Eq. (9) is the equality event defining the collision mass. Each bucket of size N_v contributes one fallback call for each of its $\binom{N_v}{2}$ unordered pairs, which proves Eq. (10); expectation then uses the pair-collision identity from Section 1. $\square$

The identity is also proved in SI Appendix, Proposition S5.1. Thus $\alpha_n(I)$ is the probability that one random comparison calls **Decide** and the expected fraction of database pairs for which this call is made; its inverse square root is the birthday scale. Figure 1 displays these identities.

3.1. Random pairs and fixed targets. Theorem 3 gives a vanishing black hole for determinant-first comparison.

Corollary 5 (Vanishing random-pair black hole). *Let D_n, D'_n be independent uniform rooted diagrams in either diagram model. Then*

$$\mathbb{P}\{(D_n, D'_n) \in \mathcal{B}_n(\det)\} = \alpha_n(\det) = O(n^{-1/2}). \quad [11]$$

Consequently, $A_{\det}$ returns a sound negative answer on a $1 - O(n^{-1/2})$ fraction of random pairs before invoking the complete procedure.

There are two distinct comparison problems. In a random-pair comparison, both determinants vary with the sample. In a fixed-target comparison, one knot K_* is prescribed and only the random diagram varies. Its determinant must then equal the fixed integer $\det K_*$. In the trefoil-slot construction, this can happen only when the number of trefoil contributions stays bounded, although its mean grows linearly with n . This yields an exponential estimate rather than the polynomial random-pair estimate.

Corollary 6 (Exponential fixed-target black hole). *For every fixed knot K_* , there are constants $C_{K_*}, c_{K_*} > 0$ such that*

$$\mathbb{P}\{\det K(D_n) = \det K_*\} \leq C_{K_*} e^{-c_{K_*} n}. \quad [12]$$

Thus determinant-first comparison with K_ invokes its complete fallback with exponentially small probability.*

The proof is given in SI Appendix, Theorem S5.3 and Corollary S5.4. If $a = \det K_*$, then

$$\det(K_n^{\text{ext}}) 3^{B_n} = a \implies B_n \leq v_3(a).$$

On $\mathcal{G}_n$, the conditional law $B_n \mid \mathcal{F}_n \sim \text{Bin}(\lfloor \delta n \rfloor, 1/4)$ has mean proportional to n , whereas $v_3(a)$ is fixed. A binomial lower-tail estimate, together with the exponentially small probability of $\mathcal{G}_n^c$, gives the fixed-target bound. The maximal-atom estimate in Theorem 3 gives the random-pair bound:

$$\text{random pair: } O(n^{-1/2}), \quad \text{fixed target: } O(e^{-cn}).$$

3.2. Refining invariants. The determinant is a coarse fingerprint. A richer invariant refines it when the richer value determines the determinant; every fiber of the richer invariant then lies inside a determinant fiber. Accordingly, call an invariant J *determinant-refining* if there is a deterministic map g with

$$\det K = g(J(K))$$

for every knot in its stated normalization. Then every J -fiber lies inside a determinant fiber, so

$$\alpha_n(J) \leq \alpha_n(\det), \quad \sup_w \mathbb{P}\{J(D_n) = w\} \leq \sup_a \mathbb{P}\{\det K(D_n) = a\}. \quad [13]$$

Hence every determinant-refining fingerprint inherits the $O(n^{-1/2})$ random-pair collision bound, and every fixed normalized value is exponentially rare. In particular, $\det K = |\Delta_K(-1)|$ in the usual normalization, so the Alexander polynomial determines the determinant and inherits both bounds. Table S4 in the SI Appendix records the required normalization conventions for the Alexander, Jones, and HOMFLY-PT polynomials, full bigraded Khovanov homology, and full bigraded knot Floer homology.

Eq. (13) compares collision probabilities, not evaluation times. The algorithmic conclusion in Corollary 5 therefore uses the determinant itself. It can be computed in polynomial bit complexity from a Goeritz matrix by fraction-free integer elimination (17, 18); the proof is in SI Appendix, Proposition S5.7. When the determinants differ, this polynomial-time branch gives a conclusive negative answer on a fraction $1 - O(n^{-1/2})$ of random pairs.

When the determinants agree, the algorithm calls a complete knot-equivalence procedure. Such a procedure exists by normal-surface methods (19, 20), and the complexity of knot and link decision problems is discussed in (21, 22). The theorem controls how often this complete branch is called. The expected and worst-case running times of the full algorithm still depend on that unrestricted procedure.

3.3. Machine learning outside and inside the black hole. A classifier can attain high accuracy by reproducing a cheap, incomplete invariant while failing to distinguish knots inside its collision fibers. This possibility is relevant to machine-learning studies of invariant prediction, conjecture discovery from learned correlations, geometric classification, and embeddings designed to be invariant under changes of diagram (23–26). Halverson and Ruehle found strong correlations between their learned embeddings and Goeritz-matrix data across the architectures they tested (26). Since the determinant is computed from a reduced Goeritz matrix, Theorem 3 and the calculation below show that a determinant-only rule already has balanced accuracy $1 - O(n^{-1/2})$ under the stated random-pair law.

Let

$$\mathcal{K}_n = \{K(D_n) \cong K(D'_n)\}, \quad q_n = \mathbb{P}(\mathcal{K}_n) < 1.$$

Consider a balanced binary benchmark whose positive class consists of equivalent pairs and whose negative class is the independent pair law conditioned on $\mathcal{K}_n^c$. The invariant-only rule

$$h_I(D, D') = \mathbf{1}_{\{I(D)=I(D')\}}$$

uses the convention that 1 means “equivalent.” Its true-positive rate is one because equivalent knots have equal invariant values. Balanced accuracy is the mean of the true-positive and true-negative rates. Here it is

$$\text{BAcc}(h_I) = 1 - \frac{1}{2} \left(\frac{\alpha_n(I) - q_n}{1 - q_n} \right) \geq 1 - \frac{\alpha_n(I)}{2}. \quad [14]$$

The identity follows from the inclusion $\mathcal{K}_n \subseteq \mathcal{B}_n(I)$ and is proved in SI Appendix, Proposition S5.9. For the normalized Alexander polynomial, $\det K = |\Delta_K(-1)|$, so Alexander equality implies determinant equality and $\alpha_n(\Delta) \leq \alpha_n(\det) = O(n^{-1/2})$. Thus this invariant-based classifier has balanced accuracy $1 - O(n^{-1/2})$. For a fixed target K_* , the corresponding balanced benchmark (positives representing K_* and negatives drawn from D_n conditioned on $K(D_n) \not\cong K_*$) has invariant-only balanced accuracy $1 - O(e^{-cn})$.

Unequal invariant values give sound negative answers, but equality leaves $\mathcal{B}_n(I)$ unresolved. Aggregate accuracy can therefore be large even when a classifier uses only information captured by polynomial invariants, an instance of shortcut learning (27). Accuracy conditional on $I(D) = I(D')$ measures what the classifier does within a collision fiber. Relevant tests use inequivalent pairs conditioned to have the same chosen invariant, including mutant and composite-knot examples (25).

3.4. Determinant comparison in independent databases. An independent database of M_n diagrams contains $\binom{M_n}{2}$ pairs. Each pair reaches the complete procedure with probability $O(n^{-1/2})$, so the expected number of fallback calls is $O(M_n^2 n^{-1/2})$. This expectation tends to zero when $M_n = o(n^{1/4})$; Markov’s inequality then shows that, with high probability, no pair reaches the complete procedure. The workload identity makes this argument precise.

Corollary 7 (Collision-free independent databases). *Let $D_n^{(1)}, \dots, D_n^{(M_n)}$ be independent uniform rooted diagrams from the complexity- n law in either diagram model. For determinant-first all-pairs comparison,*

$$\mathbb{E}F_{M_n}(\det) = O(M_n^2 n^{-1/2}), \quad [15]$$

and

$$M_n = o(n^{1/4}) \implies \mathbb{P}\{F_{M_n}(\det) > 0\} \longrightarrow 0. \quad [16]$$

Thus, with high probability, the determinant certifies every pair in such a database as inequivalent, and the complete decider is never called.

Proof. Equations (10) and (11) give Eq. (15). Markov's inequality yields

$$\mathbb{P}\{F_{M_n}(\det) > 0\} \leq \mathbb{E}F_{M_n}(\det) = O(M_n^2 n^{-1/2}),$$

which tends to zero under the stated hypothesis. $\square$

The full corollary is in SI Appendix, Corollary S5.8. The upper bound $\alpha_n(\det) = O(n^{-1/2})$ proves the no-fallback conclusion for $M_n = o(n^{1/4})$. The birthday scale is

$$M_B(n) = \alpha_n(\det)^{-1/2},$$

and the present estimate gives $M_B(n) = \Omega(n^{1/4})$. This is a sufficient collision-free range; the argument does not determine the order of $M_B(n)$.

4. Birthdays in finite knot databases

Many knot databases are fixed catalogs rather than independent samples from a random-diagram model. A census consists of a fixed list of distinct entries, and a random database of size M is then a uniformly chosen M -entry subset, sampled without replacement. This differs from Sections 2 and 3, where rooted diagrams are sampled independently from Chapman's ensembles. All formulas in this section use the finite-population law.

Let a census of T distinct entries be partitioned into nonempty invariant fibers of sizes $m_1, \dots, m_R$. For two known-distinct entries, the collision probability is

$$\alpha^{\text{neq}} = \frac{\sum_v m_v(m_v - 1)}{T(T - 1)}. \quad [17]$$

Choose a uniform M -subset of the census. Let C_M^{fp} count its equal-invariant pairs, where fp denotes finite population. Write $e_M(m_1, \dots, m_R)$ for the sum, over all choices of M distinct fibers, of the product of their sizes. This is the M th elementary symmetric polynomial.

Proposition 8 (Finite-population birthday formulas). *For $0 \leq M \leq T$,*

$$\mathbb{E}C_M^{\text{fp}} = \binom{M}{2} \alpha^{\text{neq}}, \quad \mathbb{P}\{C_M^{\text{fp}} = 0\} = \frac{e_M(m_1, \dots, m_R)}{\binom{T}{M}}. \quad [18]$$

Consequently the fiber sizes determine the no-collision probability for every M .

Proof. Every sampled pair is a uniform pair of distinct census entries, which gives the expectation. A collision-free subset chooses one entry from each of M distinct fibers. Summing the products of the chosen fiber sizes gives $e_M(m_1, \dots, m_R)$. $\square$

Further moment formulas and the without-replacement zero-one law are proved in SI Appendix, Theorem S6.1, Theorem S6.4, and Theorem S6.3. If only the census size T and the number R of occupied invariant values are known, they imply the lower bound

$$\alpha^{\text{neq}} \geq \frac{T - R}{R(T - 1)}. \quad [19]$$

At fixed T and R , balanced integer fibers minimize collision mass and maximize the no-collision probability for every M . In the synthetic 1000-entry example in SI Appendix, Proposition S7.1, the collision probability first reaches one half at $M_{50} = 17$ for a fingerprint with 900 occupied values and at $M_{50} = 38$ for a balanced fingerprint with 500 occupied values. Thus R alone does not determine the collision scale; it depends on the full fiber profile.

5. Adaptive invariant cascades

An invariant cascade evaluates a first invariant on every sampled entry and groups entries with the same value into buckets. An entry alone in its sampled bucket forms no unresolved pair and does not proceed. Only entries in buckets of size at least two pass to the next invariant. Fix a T -entry population and sample a uniform M -entry subset. After stage j , let $J_j = (I_1, \dots, I_j)$ be the joint prefix fingerprint. Two entries lie in the same J_j -bucket precisely when they agree on all of $I_1, \dots, I_j$, and $m_{J_j, v}$ denotes the corresponding population fiber size.

For any prefix J , let $S_J(M)$ be the expected number of entries that are alone in their sampled J -bucket, let $U_J(M)$ be the expected number that remain in nonsingleton buckets and pass to the next stage, and let $P_J(M)$ be the expected number of sampled pairs with equal J -values. Then

$$S_J(M) = \frac{1}{\binom{T}{M}} \sum_v m_{J, v} \binom{T - m_{J, v}}{M - 1}, \quad [20]$$

$$U_J(M) = M - S_J(M), \quad [21]$$

$$P_J(M) = \binom{M}{2} \alpha_J^{\text{neq}}. \quad [22]$$

These quantities use the joint prefix fibers and require no independence among invariant coordinates.

Proposition 9 (Expected workload of an adaptive cascade). *Let $1 \leq M \leq T$. Suppose I_j costs c_j per evaluated entry and the terminal complete procedure costs f per unresolved pair. If I_1 is evaluated on every sampled entry and each later I_j only on entries in nonsingleton sampled J_{j-1} -buckets, then*

$$\mathbb{E}W = Mc_1 + \sum_{j=2}^k c_j U_{J_{j-1}}(M) + f P_{J_k}(M). \quad [23]$$

The formula follows by linearity of expectation; details are in SI Appendix, Theorem S8.3, Proposition S8.4, and Proposition S8.5. For a fixed pair of invariants A and B , either order ends with the same joint fingerprint (A, B) and therefore the same terminal unresolved pairs. The orders differ only in how many entries require the second evaluation. Consequently,

$$A \rightarrow B \text{ is no more expensive than } B \rightarrow A \iff c_A S_B(M) \leq c_B S_A(M), \quad [24]$$

whereas the decision to append another invariant depends on its evaluation cost and the reduction in terminal pairs. Write $P_{J,I}(M)$ for the expected unresolved-pair count associated with the joint fingerprint (J, I) . Adding I after J costs $c_I U_J(M)$ in expected evaluations and saves $f[P_J(M) - P_{J,I}(M)]$ in expected terminal calls. It does not increase the expected total cost if and only if

$$c_I U_J(M) \leq f[P_J(M) - P_{J,I}(M)]. \quad [25]$$

SI Section S8 applies both inequalities to a synthetic nested population. The example is designed to separate two decisions: which of two invariants should be evaluated first, and whether adding a later invariant pays for itself. There I_2 determines I_1 , meaning that I_1 is a function of I_2 , and I_3 determines I_2 , meaning that I_2 is a function of I_3 . Thus each later invariant has finer fibers. The fibers of J_2 are consequently the fibers of I_2 , and the fibers of J_3 are the fibers of I_3 ; the same formulas apply when I_1 or I_2 is omitted.

6. Discussion

The main distinction is between random-pair separation and database uniqueness. For a discrete fingerprint with collision mass α , these are governed by α and $M^2\alpha$, respectively. The results above make this distinction precise for the Alexander polynomial and the determinant, and the collision laws extend it to every discrete invariant.

For the determinant, the source of random-pair separation is local. In each of Chapman's rooted random-diagram models there is a good-shadow event $\mathcal{G}_n$, with exponentially small complement, on which the shadow contains linearly many opened-trefoil slots. After the exterior signs are exposed, the internal signs remain conditionally independent. A single family of separating spheres gives the same exterior connected-sum factor for every internal redecoration. Each trefoil slot then contributes one factor of 3 to the determinant. Hence, on $\mathcal{G}_n$,

$$v_3(\det K(D_n)) = v_3(\det K_n^{\text{ext}}) + B_n, \quad B_n \mid \mathcal{F}_n \sim \text{Bin}(\lfloor \delta n \rfloor, 1/4).$$

On $\mathcal{G}_n$, with the shadow and exterior signs fixed, the largest conditional atom is $O(n^{-1/2})$, so the conditional collision entropy tends to infinity after the exterior crossing data have been fixed. Adding the probability of $\mathcal{G}_n^c$ gives the unconditional determinant anti-concentration estimate.

The knot argument supplies an anti-concentration estimate for one discrete law; the passage from a single comparison to a database is purely probabilistic. For two independent samples from a common discrete law, the pair-collision mass α , equivalently $H_2 = -\log \alpha$, determines the birthday scale $M_B = \alpha^{-1/2}$. For a finite population sampled without replacement, the elementary-symmetric formula gives the no-collision probability for every database size. This probability depends on the full fiber profile. At fixed T and R , balanced fibers maximize it, while imbalance can move birthday percentiles earlier. Recent finite-census studies likewise give different rankings by value count, singleton detection, and random-pair separation (12–14).

Collision probability also controls computational fallback. An invariant-first procedure gives a negative answer when the two invariant values differ and invokes a complete procedure when they agree. Thus, for two independent diagrams with law μ_n , $\alpha_n(I)$ is the probability that the comparison reaches the complete branch. In a database of independent diagrams with the same law, the number of complete calls in an all-pairs comparison is the number of colliding pairs. For two independent diagrams from either Chapman ensemble, determinant comparison gives a polynomial-time negative answer with probability $1 - O(n^{-1/2})$. The difficulty of colliding pairs and the cost of the complete procedure lie outside this conclusion.

For learned knot classifiers, collision fibers provide a controlled test. A classifier may perform well because it has recovered a familiar invariant, or because it detects information that the invariant misses. To distinguish these possibilities, it can be tested inside a common invariant fiber. Refining the fingerprint gives nested sets

$$\mathcal{B}_n(\det) \supseteq \mathcal{B}_n(\Delta) \supseteq \mathcal{B}_n(\Delta, V) \supseteq \cdots,$$

where V denotes the Jones polynomial. Let D, D' be independent with law μ_n . Whenever the conditioning event has positive probability, one may report

$$\Pi_n(f; J) = \mathbb{P}\{f(D, D') = \text{No} \mid J(D) = J(D'), K(D) \not\equiv K(D')\}.$$

This is the conditional negative accuracy for inequivalent pairs on which J agrees. It should be reported together with accuracy on equivalent pairs, since the constant-No rule has $\Pi_n(f; J) = 1$. Values of $\Pi_n(f; J)$ for different fingerprints refer to different conditioned laws and need not be monotone. Such pairs may be used as training or test examples for information not contained in J . Recent models distinguish some pairs sharing familiar invariants; $\Pi_n(f; J)$ tests this ability directly (25, 26).

The trefoil construction gives the bound $\alpha_n(\det) = O(n^{-1/2})$. For a fixed target, the hidden binomial variable is bounded by a constant, giving an exponentially small lower tail. The collision bound implies that an independent database with $M_n = o(n^{1/4})$ has no determinant collision with high probability. This argument does not determine the optimal decay rate of $\alpha_n(\det)$.

The conditional argument in SI Appendix, Theorem S3.3 gives a collision upper bound whenever its hypotheses hold. The exterior contribution is fixed under conditioning and only translates the local valuation sum, leaving its largest atom unchanged. Sharp asymptotics would require further information about the exterior contribution and its joint law with the selected block count.

The collision probability and its asymptotic rate depend on the sampling law. Chapman's ensembles are uniform on rooted diagrams of fixed crossing number; finite censuses are sampled without replacement from a fixed list; and a law on knot types weights the topology differently. Other random-knot models include lattice and off-lattice self-avoiding polygons, Petaluma diagrams, and random two-bridge knots (28–33). The collision formulas apply once the law is specified, but their values and asymptotic rates require separate analysis. In another random-knot model, an argument of the present type would require a multiplicative invariant whose valuation contains conditionally independent local summands on an event of high probability.

Finite catalogs present a further decision: which invariant should be evaluated next on entries that the preceding invariants did not separate? The cascade formulas give the expected number of entries evaluated at each stage, the unresolved-pair count, the preferred order of two invariants, and the break-even cost of another invariant, without assuming independence among the invariant coordinates. Random-pair separation, uniqueness in a database, and usefulness within a cascade are distinct properties of an invariant. The same invariant may behave quite differently in the three settings.

7. Materials and methods

7.0.0.1. Collision laws.. For fixed n , the invariant is a discrete random variable. Its atom probabilities are $(p_v)_v$, its pair-collision mass $\alpha_n = \sum_v p_v^2$ is the probability that two independent values agree, and its triple mass $\beta_n = \sum_v p_v^3$ is the probability that three agree. The variable C_M is the database collision count defined in Section 1. Its expectation and variance follow by classifying two unordered pairs of sample indices according to whether they coincide, share one index, or are disjoint. Markov's and Chebyshev's inequalities give the distribution-free zero-one law. The critical-window Poisson statement follows from factorial moments under the additional condition $M_n^3 \beta_n \rightarrow 0$. This condition controls clusters of collision pairs that share an observation. The pair-collision condition alone does not give a universal Poisson limit: a distinguished atom whose mass moves with n can preserve such clusters at the critical scale. The calculations and the counterexample exhibiting this obstruction appear in SI Appendix, Theorem S2.2, Theorem S2.3, Theorem S2.5, and Proposition S2.6.

7.0.0.2. Random diagrams and trefoil slots.. The proof separates a probabilistic statement about the underlying shadow from a topological calculation in disjoint local balls. We use the rooted general one-component and rooted reduced one-component diagram classes at fixed crossing number developed by Cantarella, Chapman, and Mastin and analyzed asymptotically by Chapman (9, 10). Forgetting the crossing information gives the corresponding rooted shadow. Every n -crossing shadow in either class has exactly 2^n decorated lifts, so a uniform diagram is obtained from a uniform rooted shadow by assigning independent fair crossing signs.

Outside an event of exponentially small probability, Chapman's connected-sum pattern theorem gives linearly many copies of a fixed asymmetric decoration of the opened-trefoil shadow. We discard the witness signs and apply a deterministic rule to the shadow, before any signs are exposed, to choose s_n occurrences at pairwise graph distance at least three. Each selected occurrence contains three crossing vertices and the five edges between them, and exactly two distinct joining edges connect it to the rest of the shadow. The separator lemma gives a shadow-measurable family of pairwise disjoint balls that works for every simultaneous internal redecoration.

We expose all exterior signs before sampling the $3s_n$ internal signs. Replacing the selected local strands by fixed trivial arcs then gives the same exterior knot K_n^{ext} for every internal sign vector. The local two-braid calculation shows that each closure is a trefoil for two of the eight decorations and an unknot for the other six. Conditional on the exposed shadow and exterior signs, the trefoil indicators are therefore independent Bernoulli(1/4) variables. These statements are proved in SI Appendix, Lemma S1.1, Lemma S4.3, Lemma S4.5, Corollary S4.6, Lemma S4.7, Lemma S4.8, Proposition S4.9, and Theorem S4.11.

7.0.0.3. Determinant and computation.. We use the symmetric Alexander polynomial with $\Delta_K(1) = 1$ and set $\det K = |\Delta_K(-1)|$. On the good-shadow event, multiplicativity under connected sum gives the valuation identity in Eq. (6). A Fourier bound for the maximal atom of $\text{Bin}(m, 1/4)$ proves the moving-target theorem (SI Appendix, Lemma S4.10 and Theorem S4.11), and a Chernoff-type lower-tail estimate gives the fixed-target exponential bound (SI Appendix, Theorem S5.3 and Corollary S5.4). For the algorithmic statements, the determinant is computed from a reduced Goeritz matrix. Fraction-free elimination and a Hadamard bound give polynomial bit complexity in the diagram size (17, 18). A complete knot-equivalence procedure makes $A_{\det}$ defined for every pair (19, 20). General complexity results for knot and link decision problems are given in (21, 22). No running-time bound for the complete branch is used here.

7.0.0.4. Finite populations and cascades.. A finite census is modeled as a fixed population of T distinct entries partitioned into invariant fibers. Databases are uniform M -subsets, so all formulas use sampling without replacement. No-collision probabilities are computed from the elementary symmetric polynomial $e_M(m_1, \dots, m_R)$ by integer dynamic programming. At fixed T and R , the extremal results follow from pairwise balancing operations on the integer fiber profile. Cascade calculations use explicit joint prefix fibers and count sampled singletons, unresolved objects, and unresolved pairs. The examples in Sections 4 and 5 are finite labeled populations specified in the SI, not empirical knot censuses. The finite-population proofs are SI Appendix, Theorem S6.1–Corollary S6.6, the cascade formulas are SI Appendix, Lemma S8.1–Proposition S8.5, and the supporting programs are available through SI Appendix, Section S9.

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7.0.0.5. Use of generative AI. During the preparation of this manuscript, the authors used ChatGPT and Codex (OpenAI; GPT-5.6 Sol and GPT-5.6 Terra through both products), Claude (Anthropic; Opus 5 and Fable 5), and Gemini (Google; Gemini 3.6 Flash and Gemini 3.1 Pro) for exploratory discussion, language editing, code review, and bibliographic checking. The research questions, mathematical ideas, theorem statements, and proofs were developed and verified by the authors. No output from these systems was treated as a mathematical source. These tools were not used to create or alter figures or research data; all figures were produced deterministically from data and $\LaTeX$ /TikZ source. The authors take full responsibility for the article.

Data, Materials, and Software Availability. The figures, supporting programs, and combined manuscript and Supporting Information PDF are available in the project repository at github.com/pedro9olivares/knots-blackholes-databases-birthdays.

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Supplementary Information for

Knots, black holes, databases, and birthdays: Collision entropy of knot invariants

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This PDF file includes:

Supplementary text
Figs. S1 to S4
Tables S1 to S7
SI References

Guide to the Supporting Information

A knot invariant assigns the same value to equivalent knots, although inequivalent knots may share that value. It can therefore be used as a fingerprint: unequal values disprove equivalence, while equal values leave the comparison unresolved. For one random pair, the unresolved probability is the collision probability of the fingerprint. A database of M objects contains $\binom{M}{2}$ pairs, so a fingerprint that usually separates one pair may still collide somewhere in a growing database. The Supporting Information treats both questions and keeps their sampling laws separate.

Four probability laws are used. Under RD, a rooted random diagram is sampled from one of Chapman’s two diagram classes. Under FT, one knot is fixed and compared with an RD object. Under DB, the database entries are independent samples. Under FC, a fixed finite census is sampled without replacement. Some estimates in Section S3 are abstract probability statements that do not require a random-diagram model; the table marks them as “abstract.” Each theorem states the law to which it applies.

Sections S2–S5 combine the collision calculation with determinant anti-concentration obtained from Chapman’s trefoil slots. Sections S6–S8 use exact fiber counts for finite-census collision probabilities and cascade workloads. The shorthand below records the logical dependencies.

The logical structure is

collision calculus $\longrightarrow$ birthday scale $\longrightarrow$ black-hole identities,
Chapman patterns $\longrightarrow$ trefoil slots $\longrightarrow$ binomial valuation coordinate,
binomial coordinate $\longrightarrow$ determinant anti-concentration,
finite fiber sizes $\longrightarrow$ collision probabilities $\longrightarrow$ adaptive workloads.

Table S1. Organization and probability laws.

Section	Subject	Results	Law
S1	Models, notation, and normalization	rooted shadow–sign product law and distinction among sampling laws	all
S2	Collisions in independent samples	collision moments, zero–one birthday law, Poisson limit under diffuseness	DB
S3	Local-block anti-concentration	lattice smoothing and conditional anti-concentration	abstract/RD
S4	Random diagrams and trefoil slots	persistent factorization, hidden binomial coordinate, determinant anti-concentration	RD
S5	Fixed targets, refinements, and computation	black-hole identities, exponential fixed-target rarity, determinant refinements, bit complexity	RD/FT/DB
S6	Collisions in a finite census	collision moments, birthday curves, balanced-fiber extremum	FC
S7	Explicit finite benchmarks	data requirements and a support–capacity example	FC
S8	Adaptive invariant cascades	survivor, pair, ordering, and break-even formulas	FC
S9	Computations and source files	repository for figures, supporting programs, and the combined PDF	all

Table S2. Main-text results and their proofs in the Supporting Information.

Main-text result	Proof in the Supporting Information
Universal inverse birthday law	Theorem S2.3, based on Theorem S2.2
Inverse birthday phenomenon	Corollary S2.4
Determinant anti-concentration	Theorem S4.11, with Lemmas S4.1–S4.3, Remark S4.4, Lemma S4.5, Corollary S4.6, Lemmas S4.7–S4.8, Proposition S4.9, and Lemma S4.10
Black-hole and workload identities	Proposition S5.1
Vanishing random-pair black hole	Corollary S5.2
Exponential fixed-target black hole	Theorem S5.3 and Corollary S5.4
Collision-free independent databases	Corollary S5.8
Finite-population birthday formulas	Theorems S6.1 and S6.4
Expected workload of an adaptive cascade	Theorem S8.3, with Propositions S8.4 and S8.5

S1. Models, notation, and normalization

A knot diagram is a plane projection of a knot together with the over–under choice at every crossing. Forgetting those choices leaves its *shadow*: a quartic planar map whose opposite-half-edge continuation records the projected components. Restoring one of the two crossing choices at every vertex turns the shadow back into a decorated diagram.

We use Chapman's rooted one-component models. A root distinguishes a map dart, and the one-component condition says that straight-ahead continuation forms a single circuit. For $\star \in \{\text{gen}, \text{red}\}$, let $\mathcal{D}_n^\star$ denote the rooted diagrams with n crossings in the general or reduced class, and let $\mathcal{S}_n^\star$ denote their shadows. The reduced condition, like the one-component condition, depends only on the shadow. This separation between the shadow and its crossing signs is essential below: the trefoil slots will be selected from the shadow before their signs are exposed.

Lemma S1.1 (Shadow–sign product law). *Every $S \in \mathcal{S}_n^\star$ has exactly 2^n decorated lifts in $\mathcal{D}_n^\star$. Consequently, a uniform $D_n \in \mathcal{D}_n^\star$ is sampled exactly by first choosing a uniform $S_n \in \mathcal{S}_n^\star$ and then assigning independent fair signs to its n crossing vertices.*

Proof. Each crossing vertex admits two over–under choices. Because the rooted shadow is fixed and the root is retained, distinct sign vectors give distinct rooted diagrams; the one-component and reduced conditions remain unchanged under decoration. Thus every admitted shadow has the same fiber size 2^n , and the uniform law factors as claimed. $\square$

Equivalently, for a rooted shadow S and sign vector $\varepsilon \in \{-1, +1\}^{V(S)}$,

$$\mathbb{P}\{S_n = S, \varepsilon_n = \varepsilon\} = \frac{1}{|\mathcal{S}_n^\star|} 2^{-n}. \quad [\text{S1.1}]$$

Sections S4–S5 concern only these rooted general and rooted reduced ensembles. No assertion is made for unrooted diagrams, for further restrictions such as graph-primality, or for a uniform distribution on knot types.

The determinant is used because it is multiplicative under connected sum, the unknot has determinant 1, and either trefoil has determinant 3. Thus each trefoil factor contributes one unit to the exponent of 3 in the determinant. For a knot K , we normalize

$$\det K := |\Delta_K(-1)|,$$

where Δ_K is the Alexander polynomial normalized by $\Delta_K(t) = \Delta_K(t^{-1})$ and $\Delta_K(1) = 1$. For a positive integer m , $v_3(m)$ is the exponent of 3 in its prime factorization. Independent copies are denoted by primes, as in D'_n . Database samples are independent unless a finite-census law is explicitly declared.

Table S3. Sampling laws used in the Supporting Information.

Code	Population	Result
RD	rooted random diagrams	asymptotic trefoil-slot theorem
FT	one fixed target plus an RD object	exponential one-atom estimate
DB	independent database	universal collision law
FC	finite census, no replacement	finite-population formulas

S2. Collisions in independent samples

At level n , let the fingerprint have a discrete law with finite or countable support. The off-critical zero–one law below requires no additional regularity; the critical window is treated separately.

S2.1. Collision mass, largest atoms, and power sums. For each n , let X_n be discrete, and let V_n denote its set of possible values. With this ambient n fixed, write

$$p_v = \mathbb{P}\{X_n = v\}, \quad v \in V_n,$$

and define

$$\alpha_n := \sum_{v \in V_n} p_v^2, \quad \beta_n := \sum_{v \in V_n} p_v^3, \quad p_n^{\max} := \sup_{v \in V_n} p_v. \quad [\text{S2.1}]$$

For an independent copy X'_n ,

$$\alpha_n = \mathbb{P}\{X_n = X'_n\}. \quad [\text{S2.2}]$$

Thus α_n is the probability of agreement in one pair, while β_n is the probability that three independent samples all agree. The latter quantity enters when two pair comparisons share one sample. The order-two Rényi entropy is

$$H_2(X_n) := -\log \alpha_n, \quad [\text{S2.3}]$$

where the natural logarithm is used unless another base is stated (1).

Lemma S2.1 (Largest atoms and power sums). *Every discrete probability law satisfies*

$$(p_n^{\max})^2 \leq \alpha_n \leq p_n^{\max}, \quad \alpha_n^2 \leq \beta_n \leq \alpha_n^{3/2}. \quad [\text{S2.4}]$$

Consequently, $\alpha_n \rightarrow 0$ if and only if $p_n^{\max} \rightarrow 0$.

Proof. For every v , $p_v^2 \leq \alpha_n$; taking the supremum gives the first lower bound. Also

$$\alpha_n = \sum_v p_v^2 \leq p_n^{\max} \sum_v p_v = p_n^{\max}.$$

For the cubic power sum, let $Z_n = p_{X_n}$. Then $\mathbb{E}Z_n = \alpha_n$ and $\mathbb{E}Z_n^2 = \beta_n$, so nonnegativity of $\text{Var}(Z_n)$ gives $\beta_n \geq \alpha_n^2$. Finally, monotonicity of ℓ^q norms gives

$$\beta_n^{1/3} = \|p\|_3 \leq \|p\|_2 = \alpha_n^{1/2}.$$

The equivalence follows from the first pair of bounds. $\square$

S2.2. Collision mean and variance. Let $X_n^{(1)}, \dots, X_n^{(M)}$ be independent copies of X_n and define

$$C_M := \sum_{1 \leq i < j \leq M} \mathbf{1}_{\{X_n^{(i)} = X_n^{(j)}\}}. \quad [\text{S2.5}]$$

The variable C_M counts colliding unordered pairs. A bucket is the set of sampled objects having one specified value. A bucket with k objects contributes $\binom{k}{2}$ to C_M . Thus nonsingleton buckets, objects in those buckets, repeated values, and colliding pairs are different statistics.

The mean locates the transition scale. A high-probability assertion above that scale requires a second-moment estimate. The cubic mass β_n appears because two collision indicators can share one database entry.

Theorem S2.2 (Independent collision formulas). *For every $M \geq 2$,*

$$\mathbb{E}C_M = \binom{M}{2} \alpha_n, \quad [\text{S2.6}]$$

$$\text{Var}(C_M) = \binom{M}{2} (\alpha_n - \alpha_n^2) + 6 \binom{M}{3} (\beta_n - \alpha_n^2). \quad [\text{S2.7}]$$

Moreover,

$$\frac{\text{Var}(C_M)}{(\mathbb{E}C_M)^2} \leq \frac{4}{M^2 \alpha_n} + \frac{4}{M \sqrt{\alpha_n}}. \quad [\text{S2.8}]$$

Proof. Write

$$I_{ij} := \mathbf{1}_{\{X_n^{(i)} = X_n^{(j)}\}}.$$

Each indicator has mean α_n , proving Eq. (S2.6). If the index pairs $\{i, j\}$ and $\{k, \ell\}$ are disjoint, then I_{ij} and $I_{k\ell}$ are independent. If two distinct index pairs share one index, for example $\{i, j\}$ and $\{i, k\}$, then

$$\mathbb{E}(I_{ij} I_{ik}) = \mathbb{P}\{X_n^{(i)} = X_n^{(j)} = X_n^{(k)}\} = \beta_n.$$

Every triple of sample indices contains three unordered pairs of index pairs sharing one index. Hence there are $3 \binom{M}{3}$ unordered overlapping pairs, and the covariance terms occur twice in the variance expansion. This proves Eq. (S2.7).

By Lemma S2.1, $\beta_n \leq \alpha_n^{3/2}$. Discarding the nonpositive terms in Eq. (S2.7) and dividing by $\binom{M}{2}^2 \alpha_n^2$ yields

$$\frac{\text{Var}(C_M)}{(\mathbb{E}C_M)^2} \leq \frac{1}{\binom{M}{2} \alpha_n} + \frac{6 \binom{M}{3}}{\binom{M}{2}^2 \sqrt{\alpha_n}}.$$

For $M \geq 2$,

$$\binom{M}{2} \geq \frac{M^2}{4}, \quad \frac{6 \binom{M}{3}}{\binom{M}{2}^2} = \frac{4(M-2)}{M(M-1)} \leq \frac{4}{M},$$

which gives Eq. (S2.8). $\square$

S2.3. The distribution-free zero-one law. Here a triangular array means only that the distribution may change with n , while the M_n samples at each fixed level are iid. Since a database of size M_n has order M_n^2 pairs, the product $M_n^2 \alpha_n$ compares the number of opportunities for a collision with the rarity of one pairwise agreement.

Theorem S2.3 (Universal independent birthday law). *For any triangular array of discrete probability laws and any integer sequence $M_n \geq 2$, let C_{M_n} denote the collision count formed from M_n iid samples from the n th law. Then:*

(i) *If $M_n^2 \alpha_n \rightarrow 0$, then*

$$\mathbb{P}\{C_{M_n} > 0\} \rightarrow 0.$$

(ii) *If $M_n^2 \alpha_n \rightarrow \infty$, then*

$$\frac{C_{M_n}}{\binom{M_n}{2} \alpha_n} \rightarrow 1 \quad \text{in probability,}$$

and therefore

$$\mathbb{P}\{C_{M_n} > 0\} \rightarrow 1.$$

No condition on the largest atom or on higher power sums is required in these two off-critical regimes.

Proof. For part (i), Markov's inequality and Eq. (S2.6) give

$$\mathbb{P}\{C_{M_n} > 0\} \leq \mathbb{E}C_{M_n} = \binom{M_n}{2} \alpha_n \rightarrow 0.$$

For part (ii), $M_n \sqrt{\alpha_n} = \sqrt{M_n^2 \alpha_n} \rightarrow \infty$, so the right-hand side of Eq. (S2.8) tends to zero. Chebyshev's inequality therefore gives

$$\frac{C_{M_n}}{\mathbb{E}C_{M_n}} \rightarrow 1 \quad \text{in probability.}$$

The mean tends to infinity. In particular,

$$\begin{aligned} \mathbb{P}\{C_{M_n} = 0\} &\leq \mathbb{P}\{|C_{M_n} - \mathbb{E}C_{M_n}| \geq \mathbb{E}C_{M_n}\} \\ &\leq \frac{\text{Var}(C_{M_n})}{(\mathbb{E}C_{M_n})^2} \rightarrow 0. \end{aligned}$$

□

Corollary S2.4 (Inverse birthday phenomenon). *Assume $\alpha_n \rightarrow 0$. Then one independent pair has different values with probability $1 - \alpha_n \rightarrow 1$. Every database sequence satisfying $M_n^2 \alpha_n \rightarrow \infty$ contains a colliding pair with probability tending to one. The scale is*

$$M_B(n) := \alpha_n^{-1/2} = \exp\left(\frac{H_2(X_n)}{2}\right). \quad [\text{S2.9}]$$

More precisely,

$$M_n = o(M_B(n)) \implies \mathbb{P}\{C_{M_n} > 0\} \rightarrow 0,$$

whereas

$$M_n/M_B(n) \rightarrow \infty \implies \mathbb{P}\{C_{M_n} > 0\} \rightarrow 1.$$

Finite- n medians and the limiting law in the critical window require further information.

S2.4. A Poisson limit in the critical window. The zero-one law leaves open the regime $M_n^2 \alpha_n = \Theta(1)$, equivalently $M_n \asymp \alpha_n^{-1/2}$. Here the expected number of collisions remains of constant order. Classical Poisson approximation results treat sparse dependent counts (2, 3). In this window, the condition $M_n^3 \beta_n \rightarrow 0$ excludes clustering at heavier atoms and yields the Poisson limit below.

Theorem S2.5 (Poisson window under diffuseness). *Suppose*

$$M_n^2 \alpha_n \rightarrow \lambda \in (0, \infty), \quad M_n^3 \beta_n \rightarrow 0. \quad [\text{S2.10}]$$

Then

$$C_{M_n} \implies \text{Poisson}(\lambda/2), \quad [\text{S2.11}]$$

and consequently

$$\mathbb{P}\{C_{M_n} > 0\} \rightarrow 1 - e^{-\lambda/2}.$$

Proof. Write $M = M_n$, let $(x)_r = x(x-1) \cdots (x-r+1)$, and set

$$s_k := \sum_v p_v^k.$$

The assumptions and $\beta_n \geq \alpha_n^2$ imply $M_n \rightarrow \infty$. Fix $r \geq 1$ and expand the falling factorial $(C_M)_r$ over ordered r -tuples of distinct edges of the complete graph on the database indices.

For one edge tuple, let G be the simple graph formed by those edges after unused vertices are deleted. If the connected-component sizes of G are $k_1, \dots, k_c$, then the probability of the corresponding equality event is

$$\prod_{j=1}^c s_{k_j}, \quad [\text{S2.12}]$$

because all sample values agree within each connected component and distinct components impose independent conditions.

If every component has two vertices, G is an r -edge matching. The number of ordered matchings is $(M)_{2r}/2^r$, so their contribution to the factorial moment is

$$\frac{(M)_{2r}}{2^r} \alpha_n^r \rightarrow \left(\frac{\lambda}{2}\right)^r.$$

Every nonmatching graph type has a component of size at least three. For $k \geq 3$, monotonicity of ℓ^q norms gives

$$s_k \leq \beta_n^{k/3}. \quad [\text{S2.13}]$$

Suppose a fixed graph type has t two-vertex components and remaining component sizes $k_1, \dots, k_u \geq 3$. Up to a constant depending only on r , the total contribution of all labeled embeddings of that type is at most

$$(M^2 \alpha_n)^t \prod_{j=1}^u (M^3 \beta_n)^{k_j/3},$$

which tends to zero by Eq. (S2.10). Only finitely many graph types occur for fixed r . Hence

$$\mathbb{E}(C_{M_n})_r \longrightarrow (\lambda/2)^r$$

for every fixed r . These are the factorial moments of $\text{Poisson}(\lambda/2)$, and the factorial-moment criterion for nonnegative integer-valued variables proves Eq. (S2.11). $\square$

S2.5. A non-Poisson critical-window example.

Proposition S2.6 (A moving heavy atom gives a non-Poisson limit). *The condition $M_n^2 \alpha_n \rightarrow \lambda$ alone does not imply a Poisson limit for C_{M_n} .*

Proof. Fix $c > 0$ and take $M_n = n$. Give one distinguished value probability c/n and distribute the remaining mass uniformly over $N_n = n^4$ additional values. Then

$$\alpha_n = \frac{c^2}{n^2} + O(n^{-4}), \quad \beta_n = \frac{c^3}{n^3} + O(n^{-8}),$$

so

$$n^2 \alpha_n \rightarrow c^2, \quad n^3 \beta_n \rightarrow c^3.$$

Let Z_n be the number of sampled copies of the distinguished value. Then $Z_n \sim \text{Bin}(n, c/n)$ and $Z_n \Rightarrow Z \sim \text{Poisson}(c)$. The expected number of collisions among the diffuse values is

$$O\left(\frac{n^2}{N_n}\right) = o(1),$$

so those collisions vanish in probability. Therefore

$$C_{M_n} \Rightarrow \binom{Z}{2}. \quad [\text{S2.14}]$$

This limit is not $\text{Poisson}(c^2/2)$: for example, $\binom{Z}{2}$ never equals 2, whereas a Poisson variable of positive mean assigns positive probability to 2. $\square$

Remark S2.7 (Collision clustering). For $\alpha_n > 0$, define

$$\kappa_n := \frac{\beta_n}{\alpha_n^{3/2}}. \quad [\text{S2.15}]$$

Lemma S2.1 gives $\sqrt{\alpha_n} \leq \kappa_n \leq 1$. In a critical window with $M_n^2 \alpha_n \rightarrow \lambda \in (0, \infty)$,

$$M_n^3 \beta_n = (M_n^2 \alpha_n)^{3/2} \kappa_n.$$

Thus the diffuseness hypothesis of Theorem S2.5 is equivalent to $\kappa_n \rightarrow 0$ in that window. The quantity α_n determines the birthday scale, while the ratio above measures clustering near the critical window.

The preceding results reduce the knot application to a uniform bound on the largest atom of a suitable invariant. The next section obtains such bounds from many independent local contributions. Its general lattice estimate includes the one-dimensional Bernoulli sum produced by trefoil slots.

S3. General local-block anti-concentration

The preceding collision calculation reduces the knot application to bounding the largest fingerprint atom, since $\alpha_n \leq p_n^{\max}$. In Section S4, a trefoil slot is a three-crossing region whose unexposed signs select either a trefoil or an unknot factor. Revealing the diagram outside linearly many such slots leaves conditionally independent choices, each contributing a Bernoulli variable of parameter $1/4$ to the 3-adic valuation of the determinant. Their sum has maximal point probability of order $n^{-1/2}$.

The estimate below records the corresponding fact for finite-support lattice variables in any dimension. It applies when, after conditioning on the revealed environment, a fingerprint coordinate is the sum of linearly many independent terms.

S3.1. Finite-support lattice smoothing. Let μ be a probability law with finite support $A \subset \mathbb{Z}^r$. Its difference lattice is

$$L_\mu := \text{span}_{\mathbb{Z}}\{a - b : a, b \in A\} \subseteq \mathbb{Z}^r.$$

The support has full affine span when L_μ has rank r .

Lemma S3.1 (Finite-support lattice smoothing). *Let $X^{(1)}, X^{(2)}, \dots$ be independent $\mathbb{Z}^r$ -valued random variables with a common finite-support law μ of full affine span. For $S_m = X^{(1)} + \dots + X^{(m)}$, there is a constant $C_\mu < \infty$ such that*

$$\sup_{z \in \mathbb{Z}^r} \mathbb{P}\{S_m = z\} \leq C_\mu m^{-r/2} \quad (m \geq 1).$$

The bound is uniform over targets $z = z_m$ that may move with m .

Proof. Let

$$\phi(t) = \mathbb{E} e^{i\langle t, X^{(1)} \rangle}, \quad t \in [-\pi, \pi]^r.$$

Fourier inversion gives

$$\mathbb{P}\{S_m = z\} = \frac{1}{(2\pi)^r} \int_{[-\pi, \pi]^r} e^{-i\langle t, z \rangle} \phi(t)^m dt, \quad [\text{S3.1}]$$

and therefore the target-independent upper bound

$$\mathbb{P}\{S_m = z\} \leq \frac{1}{(2\pi)^r} \int_{[-\pi, \pi]^r} |\phi(t)|^m dt.$$

Equality $|\phi(t)| = 1$ holds precisely when the phases $e^{i\langle t, a \rangle}$ are constant over $a \in A$, equivalently when

$$\langle t, \ell \rangle \in 2\pi\mathbb{Z} \quad \text{for every } \ell \in L_\mu.$$

Because L_μ has full rank, this condition defines a finite subgroup Γ_μ of the torus $\mathbb{R}^r/(2\pi\mathbb{Z})^r$.

The covariance matrix Σ of $X^{(1)}$ is positive definite: a zero-variance linear functional would be constant on A and would contradict full affine span. Hence, for some $c, \eta > 0$,

$$|\phi(u)| \leq e^{-c\|u\|^2} \quad \text{when } \|u\| \leq \eta.$$

For every $\gamma \in \Gamma_\mu$, the common-phase property implies $|\phi(\gamma + u)| = |\phi(u)|$. Choose disjoint η -neighborhoods of the finitely many points of Γ_μ . On their complement, continuity gives $|\phi(t)| \leq q < 1$. Consequently,

$$\int_{[-\pi, \pi]^r} |\phi(t)|^m dt \leq |\Gamma_\mu| \int_{\mathbb{R}^r} e^{-cm\|u\|^2} du + (2\pi)^r q^m \leq C m^{-r/2}.$$

Combining this estimate with Eq. (S3.1) proves the claim. $\square$

Proposition S3.2 (Sharpness of the exponent). *Fix positive probabilities $q_0, q_1, \dots, q_r$ summing to one, and let μ be supported on*

$$\{0, e_1, \dots, e_r\} \subset \mathbb{Z}^r$$

with these probabilities. Then

$$\sup_z \mathbb{P}\{S_m = z\} = \Theta(m^{-r/2}).$$

Proof. Lemma S3.1 gives the upper bound. Choose integers $n_i = mq_i + O(1)$ with $\sum_{i=0}^r n_i = m$. The point $z = (n_1, \dots, n_r)$ is reached exactly when the categorical counts are $n_0, \dots, n_r$, so

$$\mathbb{P}\{S_m = z\} = \frac{m!}{n_0! \dots n_r!} \prod_{i=0}^r q_i^{n_i}.$$

Stirling's formula gives a positive constant times $m^{-r/2}$, proving the matching lower bound. $\square$

S3.2. Conditional anti-concentration. The conditional theorem separates the revealed data from the unrevealed local choices. Here F_n is the full fingerprint, $\mathcal{E}_n$ records the environment, and $\mathcal{G}_n$ is the event on which sufficiently many blocks are available. The number of blocks is m_n , their independent contributions are $X_n^{(j)}$, and A_n contains the contribution already fixed by the environment. The map g_n selects the lattice coordinate to which the smoothing lemma is applied. Since that lemma takes a supremum over its target, the final estimate is uniform even when the most likely fingerprint value changes with n .

Theorem S3.3 (Conditional local-block anti-concentration). *For each n , let F_n be a discrete fingerprint with value set $\mathcal{V}_n$, let $\mathcal{E}_n$ be an environment sigma-field, and let $\mathcal{G}_n \in \mathcal{E}_n$ satisfy*

$$\mathbb{P}(\mathcal{G}_n^c) \leq \varepsilon_n.$$

Assume that there are:

- (i) a nonnegative $\mathcal{E}_n$ -measurable integer m_n such that $m_n \geq \delta n$ on $\mathcal{G}_n$;
- (ii) a random finite vector $\mathbf{X}_n = (X_n^{(1)}, \dots, X_n^{(m_n)})$, defined on the whole probability space, whose conditional law given $\mathcal{E}_n$ is $\mu^{\otimes m_n}$ almost surely on $\mathcal{G}_n$, where μ is one fixed finite-support law on $\mathbb{Z}^r$ of full affine span;
- (iii) an $\mathcal{E}_n$ -measurable remainder $A_n \in \mathbb{Z}^r$; and
- (iv) a deterministic map $g_n : \mathcal{V}_n \rightarrow \mathbb{Z}^r$ such that, on $\mathcal{G}_n$,

$$g_n(F_n) = A_n + \sum_{j=1}^{m_n} X_n^{(j)}.$$

Then, for a constant C depending only on μ and δ ,

$$\sup_{v \in \mathcal{V}_n} \mathbb{P}\{F_n = v\} \leq \varepsilon_n + Cn^{-r/2}.$$

The estimate is uniform over moving fingerprint values $v = v_n$.

Proof. Fix $v \in \mathcal{V}_n$. Since $\mathcal{G}_n \in \mathcal{E}_n$,

$$\mathbb{P}\{F_n = v, \mathcal{G}_n\} = \mathbb{E}[\mathbf{1}_{\mathcal{G}_n} \mathbb{P}\{F_n = v \mid \mathcal{E}_n\}].$$

On $\mathcal{G}_n$, the event $\{F_n = v\}$ is contained in

$$\left\{ \sum_{j=1}^{m_n} X_n^{(j)} = g_n(v) - A_n \right\}.$$

Because m_n and A_n are $\mathcal{E}_n$ -measurable, the conditional product law and Lemma S3.1 give

$$\mathbf{1}_{\mathcal{G}_n} \mathbb{P}\{F_n = v \mid \mathcal{E}_n\} \leq \mathbf{1}_{\mathcal{G}_n} C_\mu m_n^{-r/2} \leq C_\mu \delta^{-r/2} n^{-r/2}.$$

Taking expectations and adding $\mathbb{P}(\mathcal{G}_n^c)$ proves the result. □

Remark S3.4. No independence is required between A_n and the number or positions of the blocks. The blocks are selected before their internal random variables are exposed. For the trefoil argument, $r = 1$, $X_n^{(j)} \sim \text{Bernoulli}(1/4)$, and $g_n = v_3 \circ \det$. There $m_n = s_n = \lfloor \delta n \rfloor$, so hypothesis (i) holds with density constant $\delta/2$ for all sufficiently large n .

S4. Random diagrams, trefoil slots, and determinant anti-concentration

For either rooted Chapman ensemble, the maximal determinant atom is $O(n^{-1/2})$. We prove this by exhibiting a conditional binomial coordinate in $v_3(\det K(D_n))$. Chapman's pattern theorem first gives linearly many disjoint opened-trefoil shapes outside an exponentially small exceptional set (4). These shapes are selected from the shadow, before their crossing signs are exposed. They can therefore be thinned to linearly many well-separated slots without changing the independent fair law of their internal signs.

The topological step places the selected slots in pairwise disjoint balls that remain valid under every simultaneous choice of the unexposed signs. Each slot then contributes a connected-sum factor. Of its eight equally likely sign triples, two close to a trefoil and six to the unknot. The trefoil count is therefore conditionally binomial with parameter $1/4$ and, added to $v_3(\det K_n^{\text{ext}})$, gives $v_3(\det K(D_n))$. The largest atom of this binomial variable gives the required bound.

structured tangle disk Δ

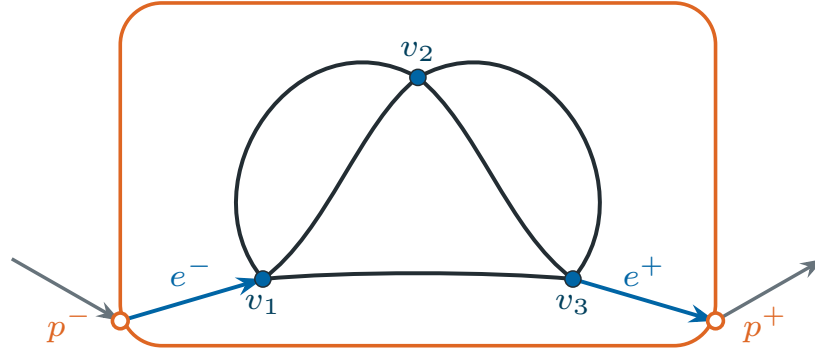


Fig. S1. The opened three-crossing trefoil shadow Q . Closing its two exterior legs restores the doubled triangle, the standard shadow of the closure of the two-braid σ_1^3 . The structured pattern retains the tangle disk, the marked tail, and the two joining sides.

S4.1. The opened-trefoil pattern. Let Q be the two-leg shadow obtained by cutting one closure edge of the standard three-crossing two-braid shadow. We retain the tangle disk, a marked tail, and the ordered joining sides, so an occurrence includes the data needed by Chapman's edge-replacement construction. Label the three crossing vertices v_1, v_2, v_3 from tail to head.

Lemma S4.1 (Admissibility of the opened trefoil). *The structured shadow Q has one open strand and no closed strand, is prime and two-leg-prime, is reduced, and admits an asymmetric decoration. Hence it satisfies the local hypotheses for Chapman's viable connect-sum attachment, pattern theorem, and reduced diagram class.*

Proof. Following straight-ahead continuation from the tail passes through all three crossing vertices and exits at the head; every crossing dart belongs to this open strand, so there is no closed component. Closing the two legs produces the doubled triangle. This multigraph remains connected after deletion of any two edges, so Q is prime in Chapman's graph-theoretic sense. In particular, deleting the cut closure edge and any one additional edge leaves it connected, which verifies two-leg-primality. In the doubled-triangle rooted closure, deleting any crossing vertex leaves the other two joined by two parallel edges. Hence the rooted closure is reduced; equivalently, the opened tangle has no disconnecting internal crossing vertex.

An orientation-preserving automorphism fixing both exterior legs fixes the connected rooted tangle dart by dart and is the identity. Any remaining automorphism must exchange the legs and therefore exchange v_1 with v_3 . Decorate v_1 negatively and v_2, v_3 positively. This decoration is not preserved by the exchange, so it is asymmetric. Denote the resulting fixed decorated tangle by P . Graph-theoretic primeness is used in Chapman's Proposition 10, two-leg-primality and asymmetry in the viable attachment of Proposition 3, and reducedness in the reduced diagram class. The fixed signs of P are used to invoke Chapman's theorem and are forgotten before the crossings are randomly redecorated. $\square$

S4.2. Chapman's pattern theorem. For a decorated diagram D , let $\nu_P(D)$ be the maximum number of pairwise vertex-disjoint structured occurrences of P as connect-summand tangles. Chapman's Proposition 3 verifies viable edge replacement for the two-leg-prime asymmetric tangle P ; Theorems 6 and 9 provide the required smooth-growth pattern theorem, and Proposition 10 applies it to this prime tangle in the rooted general and rooted reduced diagram classes (4). Taking the smaller abundance constant and the larger exceptional-rate constant for the two classes gives common $c > 0$ and $d \in (0, 1)$ such that, for either class and all sufficiently large n ,

$$\mathbb{P}\{\nu_P(D_n) < cn\} \leq d^n. \quad [\text{S4.1}]$$

The occurrences counted by the viable attachment are pairwise vertex-disjoint.

S4.3. Passing from diagrams to shadows. For a rooted shadow S , let $\nu_Q(S)$ be the maximum number of pairwise vertex-disjoint structured Chapman occurrences of Q : contained tangles in the sense of (4), with the pattern-side designation, tangle disk, marked tail, and ordered joining sides retained. If $\pi D = S$, every structured P -occurrence in D becomes an occurrence of this class in S , so

$$\nu_P(D) \leq \nu_Q(\pi D).$$

Lemma S4.2 (Decorated-to-shadow transfer). *With c and d as in Eq. (S4.1), for every fixed $\eta \in (0, c)$ and all sufficiently large n ,*

$$\mathbb{P}\{\nu_Q(S_n) < \eta n\} \leq d^n,$$

where $S_n = \pi(D_n)$ is the uniform rooted shadow obtained from either the rooted general or the rooted reduced diagram class.

Proof. Let

$$A_n = \{S \in \mathcal{S}_n^* : \nu_Q(S) < \eta n\}.$$

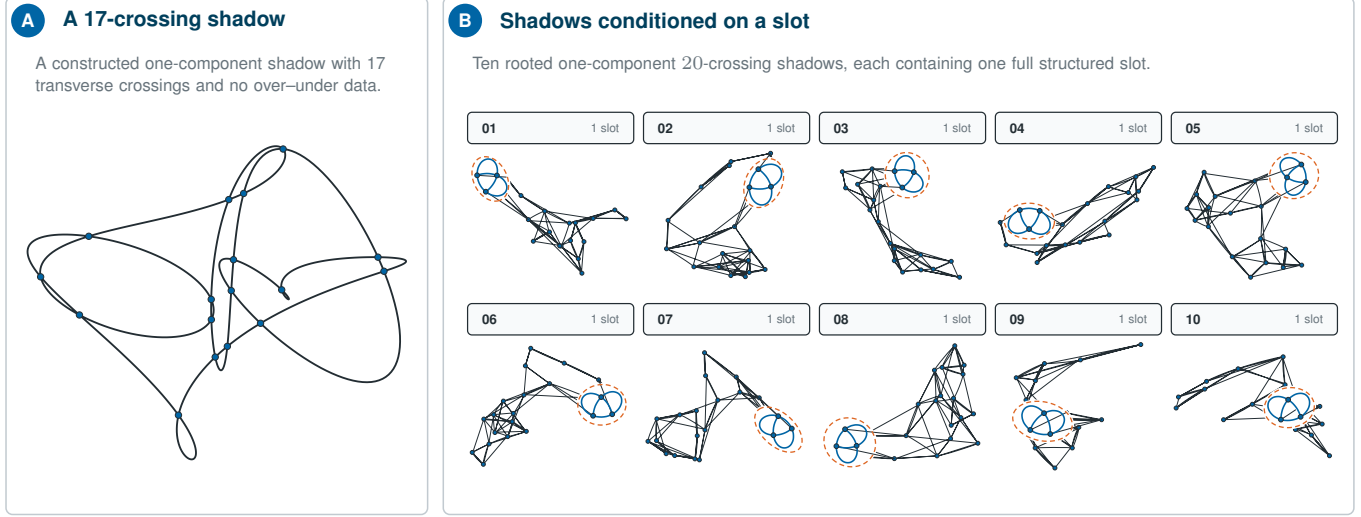


Fig. S2. Finite shadow examples. (A) A constructed one-component shadow with 17 transverse crossings. (B) Ten rooted one-component 20-crossing shadows from a deterministic PlanarMap search conditioned on the presence of a full structured opened-trefoil slot. Of 454 uniform quartic-map proposals, 82 were one-component; every displayed shadow has exactly one such slot, marked by a dashed orange circle. The planar isotopies place the slots in the standard opened-trefoil position while preserving incidence and straight-ahead continuation. Blue dots mark degree-four shadow vertices, and neither panel assigns over-under information. These conditioned examples illustrate the local pattern; Chapman's theorem supplies the asymptotic statement (4).

If $S \in A_n$, then every decorated lift D of S satisfies $\nu_P(D) \leq \nu_Q(S) < cn$. Hence

$$\pi^{-1}(A_n) \subseteq \{D : \nu_P(D) < cn\}.$$

By Eq. (S1.1), all shadows have the same number 2^n of lifts, so the projection of the uniform diagram law is the uniform shadow law. Therefore

$$\mathbb{P}\{S_n \in A_n\} = \mathbb{P}\{D_n \in \pi^{-1}(A_n)\} \leq \mathbb{P}\{\nu_P(D_n) < cn\} \leq d^n.$$

□

S4.4. Choosing well-separated occurrences. All occurrences considered below are structured Chapman occurrences in the preceding sense. Each is encoded by finitely many darts in the rooted planar map. Order all such encodings lexicographically. Among all maximum vertex-disjoint Q -families, choose the lexicographically first. This choice is a deterministic function of the shadow.

For two occurrences, define their distance to be the graph distance between their vertex sets in the four-valent shadow. Starting with the chosen maximum family, scan its members in lexicographic order. Retain the first remaining occurrence and delete every other occurrence at distance at most two. Repeat until no occurrence remains. Fix $\eta \in (0, c)$ and set $\mathcal{G}_n = \{\nu_Q(S_n) \geq \eta n\}$.

Lemma S4.3 (Shadow-measurable distance-three selection). *There are constants $\delta > 0$, $C_0 < \infty$, and $c_0 > 0$, and a deterministic shadow-measurable rule such that, outside an event of probability at most $C_0 e^{-c_0 n}$, the rule selects exactly*

$$s_n = \lfloor \delta n \rfloor$$

structured Q -slots whose vertex sets have pairwise graph distance at least three.

Proof. For a retained occurrence H , let $N_2(V(H))$ be the closed radius-two neighborhood of its three internal vertices. A nonbacktracking exploration in the four-valent multigraph gives

$$|N_2(V(H))| \leq 3 + 3 \cdot 4 + 3 \cdot 4 \cdot 3 = 51;$$

loops, parallel edges, and repeated vertices only lower this count. Charge each occurrence deleted with H , and H itself, to H . Every charged occurrence meets $N_2(V(H))$, and vertex-disjointness of the starting family makes chosen meeting vertices distinct. Thus at most 51 starting occurrences are charged to H , and the thinning retains at least $\nu_Q(S_n)/51$ occurrences.

Choose $\delta \in (0, \eta/51)$. On $\mathcal{G}_n$ the first $s_n = \lfloor \delta n \rfloor$ retained occurrences exist and have pairwise distance at least three. Taking these when they exist, and the entire retained family otherwise, defines a deterministic shadow-measurable rule. Lemma S4.2 gives $\mathbb{P}(\mathcal{G}_n^c) \leq d^n$ for all sufficiently large n ; take $c_0 = -\log d$ and enlarge C_0 to absorb the finitely many remaining values of n . □

Remark S4.4 (Order of exposure). The selection in Lemma S4.3 is made from S_n alone. Conditional on S_n and on all crossing signs outside the selected slots, the $3s_n$ internal signs are therefore still independent fair signs. Their conditional law remains a product law because the slots are chosen before any of these signs is exposed.

S4.5. Trefoil decorations and simultaneous factorization. On $\mathcal{G}_n$, the selected slots are determined by the shadow. The separating disks constructed below depend only on that shadow, are fixed before any crossing signs in the slots are exposed, and work for every simultaneous internal redecoration. We then identify the local two-braid closures and the resulting factorization for every internal sign vector.

S4.6. The two-edge boundary. To isolate a slot by a sphere meeting the knot in exactly two points, its two boundary incidences must lie on distinct ambient edges. Chapman's general submap definition allows a boundary walk to traverse an edge more than once, so boundary length two alone does not provide this fact. For the opened-trefoil pattern, the valence count at the three internal crossings proves that the two joining edges are distinct.

Lemma S4.5 (Boundary of an opened-trefoil occurrence). *Let H be a structured Chapman occurrence of the opened-trefoil shadow Q in a connected four-valent shadow S with more than three vertices. Write $V(H)$ for its three internal crossing vertices, let H_{int} be the image of the five internal edges of Q , and let e_H^- and e_H^+ be the joining edges determined by the marked tail and the ordered joining sides. Then*

$$S[V(H)] = H_{\text{int}}, \quad \delta_S(V(H)) = \{e_H^-, e_H^+\}.$$

The two joining edges are distinct and each has exactly one endpoint in $V(H)$. There are no other edges of S between vertices of $V(H)$ and no other edges leaving $V(H)$. The structured occurrence also transports the cyclic orders, hence the straight-ahead pairing and the unique open strand of Q , together with its tangle disk and pattern side. In particular, the marked exterior face of Q^ is bounded by two distinct dual edges.*

Proof. The internal multigraph of Q has two edges from v_1 to v_2 , two from v_2 to v_3 , and one from v_3 to v_1 . Its five internal edges therefore use ten darts at the three internal crossings. The tail leg at v_1 and the head leg at v_3 use the remaining two darts. Altogether the structured tangle accounts for

$$2|E_{\text{int}}(Q)| + 2 = 2 \cdot 5 + 2 = 12$$

incident darts.

Since S is four-valent and $|V(H)| = 3$,

$$\sum_{v \in V(H)} \deg_S(v) = 12.$$

The five transported internal edges exhaust ten of these ambient darts, so exactly two darts incident with $V(H)$ remain: the transported tail and head legs. These two darts cannot be paired to each other. If they were, every edge incident with $V(H)$ would have both endpoints in $V(H)$; because H_{int} is connected, $V(H)$ would then be the vertex set of a connected component of S , contrary to the connectedness of S and $|V(S)| > 3$. Thus each leg dart is paired with a dart outside $V(H)$. Edge pairings are disjoint, so the resulting joining edges e_H^- and e_H^+ are distinct. The count now proves

$$S[V(H)] = H_{\text{int}}, \quad \delta_S(V(H)) = \{e_H^-, e_H^+\}.$$

Finally, an isomorphism of oriented plane maps preserves the cyclic order of the four darts at each crossing. Opposite darts in that cyclic order are the straight-ahead mates, so the straight-ahead transition system on H is the one on Q . Lemma S4.1 identifies it as a single interval from the marked tail to the marked head, with no closed strand. The retained tangle disk determines the pattern side, and the marked tail and ordered joining sides retain the remaining structured data. The two distinct joining edges dualize to the two distinct boundary edges of the marked exterior 2-gon of Q^* . $\square$

Corollary S4.6 (Separation of the joining edges). *On $\mathcal{G}_n$, let V_j be the internal vertex set of the j th selected slot. If $i \neq j$, no joining edge of the i th slot ends in V_j , and joining edges from the i th and j th slots have no common exterior endpoint. Hence cut points may be chosen near their internal endpoints so that the truncated subgraphs are pairwise disjoint; after collapse and suppression of the cut points, the marked residual edges are pairwise vertex-disjoint. The same values of s_n , δ , C_0 , and c_0 apply.*

Proof. The two forbidden incidences would give paths of length one and two between V_i and V_j , contradicting Lemma S4.3. Lemma S4.5 identifies the joining edges as all edges leaving each selected subgraph. Distinct plane-map edges have disjoint interiors, so short initial joining-edge segments give disjoint truncations. After collapse, a shared endpoint of two marked residual edges would be a shared exterior endpoint of joining edges from the corresponding slots. The selected family is unchanged. $\square$

S4.7. Disjoint tangle disks for all redecorations. Graph-distance separation makes the truncated slot subgraphs disjoint. Connected-sum factorization additionally requires planar disks with disjoint interiors whose boundaries meet the shadow exactly twice. These disks must be chosen before the internal signs are known, so that one family works for every redecoration. The next lemma supplies this separation.

Lemma S4.7 (Pairwise disjoint separating disks). *On the good-shadow event $\mathcal{G}_n$, let $H_1, \dots, H_{s_n}$ be the selected structured Q -slots in the one-component shadow $S = S_n$. Write $V_j = V(H_j)$ and let e_j^- and e_j^+ be the joining edges specified by the marked tail and ordered joining sides. There are pairwise disjoint closed disks*

$$\Delta_1, \dots, \Delta_{s_n} \subset S^2$$

and points $p_j^- \in \text{int}(e_j^-)$ and $p_j^+ \in \text{int}(e_j^+)$ with the following properties.

(i) The boundary $\partial\Delta_j$ contains no vertex of S and meets S transversely exactly at p_j^- and p_j^+ .

(ii) Let $u_j^\pm \in V_j$ be the internal endpoint of $e_j^\pm$, and let

$$T_j = S[V_j] \cup [u_j^-, p_j^-]_{e_j^-} \cup [u_j^+, p_j^+]_{e_j^+}.$$

Then $\Delta_j \cap S = T_j$. Thus Δ_j contains precisely the selected opened-trefoil subgraph and the indicated initial joining-edge segments.

(iii) Replacing every T_j inside Δ_j by a properly embedded arc from p_j^- to p_j^+ , and suppressing the resulting bivalent subdivision points, produces a planar one-component residual shadow S° . The corresponding marked residual edges $f_1, \dots, f_{s_n}$ are pairwise vertex-disjoint. Reinserting the retained structured occurrences recovers S ; abstract copies with the same recorded data reconstruct a shadow orientation-preservingly isomorphic to S by a homeomorphism that is the identity outside $\bigcup_j \Delta_j$.

(iv) The disks use only the structured shadow data. Once the shadow is fixed, the same disks and a fixed family of pairwise disjoint 3-ball thickenings work for every simultaneous redecoration of the selected crossings. Each boundary sphere meets every such decorated knot in exactly two points.

Proof. The assertion is empty when $s_n = 0$, so suppose $s_n \geq 1$. Lemma S4.5 shows that each selected opened-trefoil subgraph is induced by V_j and that

$$\delta_S(V_j) = \{e_j^-, e_j^+\}.$$

Lemma S4.3 gives $\text{dist}_S(V_i, V_j) \geq 3$ for $i \neq j$.

Connectedness of the exterior. Fix j , and let G_j^{out} be the subgraph of S induced by $V(S) \setminus V_j$. We first show that G_j^{out} is connected. Let C be one of its connected components. Since S is connected, at least one edge joins C to V_j . No edge joins C to another component of G_j^{out} , so every edge of $\delta_S(V(C))$ belongs to $\delta_S(V_j)$. All vertices of S have even degree, and hence

$$|\delta_S(V(C))| \equiv \sum_{v \in V(C)} \deg_S(v) \equiv 0 \pmod{2}.$$

Every component of G_j^{out} is therefore incident with a positive even number of the two joining edges. There can be only one component.

Disjoint truncated subgraphs. Write $w_j^\pm \in V(S) \setminus V_j$ for the exterior endpoint of $e_j^\pm$. Choose $p_j^\pm \in \text{int}(e_j^\pm)$ sufficiently close to $u_j^\pm$ and define T_j as in the statement. Corollary S4.6 permits these choices so that the compact sets T_j are pairwise disjoint. Set

$$X^{(j)} = G_j^{\text{out}} \cup [p_j^-, w_j^-]_{e_j^-} \cup [p_j^+, w_j^+]_{e_j^+}.$$

The two-edge identities give

$$S = T_j \cup X^{(j)}, \quad T_j \cap X^{(j)} = \{p_j^-, p_j^+\}.$$

The connectedness of G_j^{out} shows that $X^{(j)}$ and $X^{(j)} \setminus \{p_j^-, p_j^+\}$ are connected.

Individual disks from ribbon neighborhoods. Choose a closed ribbon neighborhood N_j of T_j , cut by small transverse cross-arcs at p_j^- and p_j^+ , so that

$$N_j \cap S = T_j, \quad \partial N_j \cap S = \{p_j^-, p_j^+\}.$$

Because the T_j are pairwise disjoint and finite in number, the neighborhoods N_j may be chosen pairwise disjoint. Each N_j is a connected compact genus-zero subsurface of S^2 , so every component of $S^2 \setminus N_j$ is an open disk. The connected set $X^{(j)} \setminus \{p_j^-, p_j^+\}$ is disjoint from N_j and hence lies in a unique component U_j of $S^2 \setminus N_j$. Both cut points lie on ∂U_j , since the two exterior edge segments approach them. Define

$$\Delta_j = S^2 \setminus U_j.$$

Since U_j is an open disk, Δ_j is a closed disk. Its boundary is contained in ∂N_j , contains both cut points, and meets the shadow nowhere else. The intersections at the cut points are transverse. Moreover, $T_j \subset \Delta_j$, while every point of $S \setminus T_j$, apart from the two cut points, belongs to $X^{(j)} \setminus \{p_j^-, p_j^+\} \subset U_j$. Thus

$$\Delta_j \cap S = T_j.$$

Disjointness of the disks. The curves $\partial\Delta_j$ are pairwise disjoint because $\partial\Delta_j \subset \partial N_j$ and the N_j are pairwise disjoint. It remains to rule out incompatible choices of disk side. Choose a point x on the open segment of e_1^- from p_1^- toward w_1^- , avoiding all cut points. No other T_j contains an interior point of this edge: otherwise that edge would be internal to, or a joining edge of, another selected occurrence, producing distance at most one between the corresponding internal vertex sets. Hence

$$x \in X^{(j)} \setminus \{p_j^-, p_j^+\} \subset U_j \quad \text{for every } j,$$

so $x \notin \Delta_j$ for every j .

Identify $S^2 \setminus \{x\}$ with $\mathbb{R}^2$. Each Δ_j is now the bounded closed Jordan domain determined by $\partial\Delta_j$. Two such domains with disjoint boundaries are disjoint or one lies in the interior of the other. If $\Delta_k \subset \text{int}(\Delta_j)$ for $j \neq k$, then $T_k \subset \Delta_j$. On the other hand, the pairwise disjointness of the truncated subgraphs and the decomposition $S = T_j \cup X^{(j)}$ give

$$T_k \subset X^{(j)} \setminus \{p_j^-, p_j^+\} \subset U_j = S^2 \setminus \Delta_j,$$

a contradiction. Thus the disks are pairwise disjoint.

The residual shadow. For every j , choose a properly embedded arc $\gamma_j \subset \Delta_j$ from p_j^- to p_j^+ , with interior in $\text{int}(\Delta_j)$ and with endpoint germs compatible with the two joining-edge segments. Remove $T_j \setminus \{p_j^-, p_j^+\}$ and insert γ_j , for all j simultaneously. Suppressing the bivalent points p_j^- and p_j^+ produces a planar four-valent residual shadow S° and a marked residual edge f_j for every collapsed occurrence.

The shadow S° still has one straight-ahead component. Lemma S4.5 identifies the transition system inside T_j as the single interval strand from p_j^- to p_j^+ , with no closed strand. Replacing this interval by γ_j preserves the continuation of the unique global straight-ahead circuit. The disjointness of the disks makes all replacements simultaneous. Corollary S4.6 gives that the marked residual edges are pairwise vertex-disjoint.

Reinserting the retained embedded copies of the T_j reverses the construction and recovers S . If the operation is encoded using abstract copies, the marked tail, ordered joining sides, pattern side, and rotation system identify each abstract tangle with the retained one relative to the two cut points. Extending this plane-tangle identification across the complementary disks of Δ_j and correcting its boundary restriction by an isotopy gives an orientation-preserving homeomorphism of Δ_j fixed on $\partial\Delta_j$. The disk homeomorphisms therefore glue to the identity outside $\bigcup_j \Delta_j$.

Ball thickenings independent of the signs. All cut points, ribbon neighborhoods, exterior components, and disks were chosen from the structured shadow before any selected crossing sign was exposed. Fix a product neighborhood $S^2 \times [-\epsilon, \epsilon]$. Choose disjoint crossing cylinders at every crossing of S , with their projection disks disjoint from every $\partial\Delta_j$. Use the same cylinders and absolute crossing height h , with $0 < h < \epsilon$, in every realization. Keep the exterior crossing choices fixed and vary only the crossings inside the selected disks. Away from the union of all crossing cylinders, the lifted knot lies in $S^2 \times \{0\}$.

For each j , take a rounded product thickening of $\Delta_j \times [-\epsilon, \epsilon]$, leaving its side boundary unchanged near p_j^- and p_j^+ . It is a 3-ball whose boundary misses the lifted knot on the upper and lower faces and meets it transversely on the side boundary exactly at the two cut points. The planar disks are pairwise disjoint, so the product thickenings may be chosen pairwise disjoint. They are fixed before the signs are chosen and therefore work for every simultaneous internal redecoration. $\square$

S4.8. The eight local decorations. Each selected slot has three unexposed crossing signs and therefore eight equally likely decorations. We enumerate them to determine the local knot factor and to identify the exact Bernoulli variable that enters the determinant.

Work on $\mathcal{G}_n$, and write $H_1, \dots, H_{s_n}$ for the selected slots. For each such rooted shadow, choose one family of disks, ball thickenings, and residual arcs given by Lemma S4.7. At fixed n there are finitely many rooted shadows, so the choice may be made shadow-measurably.

For every j , transport the labels $v_{j,1}, v_{j,2}, v_{j,3}$ and the reference decoration from the standard cut-open $\hat{\sigma}_3^3$ model. Let $\varepsilon_{n,j,r} = +1$ when the sign at $v_{j,r}$ agrees with the transported reference crossing and let $\varepsilon_{n,j,r} = -1$ when it is switched. Put

$$\varepsilon_{n,j} = (\varepsilon_{n,j,1}, \varepsilon_{n,j,2}, \varepsilon_{n,j,3}), \quad \varepsilon = (\varepsilon_{n,1}, \dots, \varepsilon_{n,s_n}),$$

and let $\mathcal{F}_n$ be the sigma-field generated by S_n and all crossing signs outside the selected internal vertex sets. Give ε a fixed value on $\mathcal{G}_n^c$, so that it is defined everywhere. The selected list and $\mathcal{G}_n$ are shadow-measurable, hence $\mathcal{G}_n \in \mathcal{F}_n$. The shadow-sign product law gives

$$\mathcal{L}(\varepsilon \mid \mathcal{F}_n) = \text{Unif}((\{-1, +1\}^3)^{s_n}) \quad \text{almost surely on } \mathcal{G}_n. \quad [\text{S4.2}]$$

Lemma S4.8 (Closure of a decorated opened trefoil). *Once a generator σ_1 of B_2 is fixed, there is a sign $\eta_H \in \{-1, +1\}$, determined by the structured shadow embedding and independent of the decoration, such that the recorded closure of the decorated slot is*

$$\sigma_1^{\eta_H \varepsilon_1} \widehat{\sigma_1^{\eta_H \varepsilon_2}} \sigma_1^{\eta_H \varepsilon_3} \cong T(2, \eta_H(\varepsilon_1 + \varepsilon_2 + \varepsilon_3)).$$

The local connected-sum factor defined by a separator ball from Lemma S4.7 has this isotopy type. Thus the two unanimous triples give the trefoil mirror pair, and every mixed triple gives the unknot.

Proof. The retained tangle disk, marked tail, ordered joining sides, and rotation system identify H with the standard three-crossing two-braid shadow cut open along one closure edge. Relative to the chosen generator, the transported reference crossing at all three braid positions is $\sigma_1^{\eta_H}$ for one common sign η_H ; switching a crossing gives its inverse. Since $B_2 = \langle \sigma_1 \rangle \cong \mathbb{Z}$, the exponents add, and the standard closure of σ_1^k is $T(2, k)$.

A separator ball contains exactly the structured one-strand tangle and meets the knot at its two joining points. The structured identification extends across the two joining-edge collars to the cut points. Under this identification, the missing closure edge of the standard model is represented by a boundary-parallel cap in the complementary ball used to close the local tangle. Any two such caps are ambient isotopic relative to their endpoints, so this cap computes the factor defined by the separating sphere. The resulting local knot is the displayed braid closure. Its exponent is ± 3 for a unanimous triple and ± 1 otherwise; $T(2, \pm 3)$ is the trefoil mirror pair and $T(2, \pm 1)$ is the unknot. $\square$

On $\mathcal{G}_n$, conditional on $\mathcal{F}_n$, the eight sign triples at each slot are equally likely. Write $s = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$; the braid exponent is $\eta_H s$.

$(\varepsilon_1, \varepsilon_2, \varepsilon_3)$	s	closure	$(\varepsilon_1, \varepsilon_2, \varepsilon_3)$	s	closure
$(+, +, +)$	3	trefoil	$(-, -, -)$	-3	trefoil
$(+, +, -)$	1	unknot	$(-, -, +)$	-1	unknot
$(+, -, +)$	1	unknot	$(-, +, -)$	-1	unknot
$(-, +, +)$	1	unknot	$(+, -, -)$	-1	unknot

Define

$$Y_{n,j} := \mathbf{1}\{\varepsilon_{n,j,1} = \varepsilon_{n,j,2} = \varepsilon_{n,j,3}\}.$$

Lemma S4.8 shows that $Y_{n,j} = 1$ when the j th local factor is a trefoil and $Y_{n,j} = 0$ when it is an unknot. Eq. (S4.2) therefore gives

$$\mathcal{L}((Y_{n,1}, \dots, Y_{n,s_n}) \mid \mathcal{F}_n) = \text{Bernoulli}(1/4)^{\otimes s_n} \quad \text{almost surely on } \mathcal{G}_n. \quad [\text{S4.3}]$$

The mirror distinction is irrelevant for the determinant because both trefoils have determinant 3.

S4.9. Connected-sum factorization and the 3-adic valuation. For $\mathbf{e} = (\mathbf{e}_1, \dots, \mathbf{e}_{s_n}) \in (\{-1, +1\}^3)^{s_n}$, let $D_n[\mathbf{e}]$ denote the diagram obtained from the exposed shadow and exterior signs by assigning the reference-relative sign triple $\mathbf{e}_j$ to the j th selected slot.

Proposition S4.9 (Simultaneous connected-sum factorization). *On $\mathcal{G}_n$, there are an $\mathcal{F}_n$ -measurable knot type K_n^{ext} and $\mathcal{F}_n$ -measurable maps*

$$L_{n,j} : \{-1, +1\}^3 \longrightarrow \{\text{knot types}\}, \quad 1 \leq j \leq s_n,$$

such that, simultaneously for every internal sign vector $\mathbf{e}$,

$$K(D_n[\mathbf{e}]) \cong K_n^{\text{ext}} \# L_{n,1}(\mathbf{e}_1) \# \dots \# L_{n,s_n}(\mathbf{e}_{s_n}). \quad [\text{S4.4}]$$

The knot K_n^{ext} is the same for all $\mathbf{e}$. Moreover, $L_{n,j}(\mathbf{e}_j)$ is an unknot for a mixed triple and is the appropriate trefoil chirality for a unanimous triple.

Proof. Use the family of pairwise disjoint separator balls fixed above. In the j th ball, lift the fixed residual arc from Lemma S4.7 to a boundary-parallel trivial arc τ_j with the same two boundary points. For any $\mathbf{e}$, replace the decorated local strand inside every ball by τ_j . All dependence on $\mathbf{e}$ lies inside the balls, so the resulting decorated residual diagram is the same for every $\mathbf{e}$. Its knot type is K_n^{ext} . The residual shadow and the crossing signs retained outside the balls are $\mathcal{F}_n$ -measurable, hence so is K_n^{ext} .

For each j , close the decorated local tangle in an abstract complementary ball by a boundary-parallel cap and call the resulting knot $L_{n,j}(\mathbf{e}_j)$. The structured embedding and the fixed cap are determined by the exposed shadow, so the finite map $L_{n,j}$ is $\mathcal{F}_n$ -measurable. Cutting along the disjoint boundary spheres and capping the two sides is the standard connected-sum operation. Applying it successively to the pairwise disjoint spheres gives Eq. (S4.4) for every $\mathbf{e}$ with the same exterior factor K_n^{ext} . Lemma S4.8 identifies the local closure as the unknot for a mixed triple and as a trefoil for a unanimous triple. $\square$

For the random internal vector ε , set

$$B_n = \sum_{j=1}^{s_n} Y_{n,j}.$$

Extend K_n^{ext} arbitrarily to $\mathcal{G}_n^c$; since $\mathcal{G}_n \in \mathcal{F}_n$, it remains $\mathcal{F}_n$ -measurable. The fixed extension of ε also defines B_n there. Proposition S4.9, determinant multiplicativity, and Eq. (S4.3) give, almost surely on $\mathcal{G}_n$,

$$\det K(D_n) = \det(K_n^{\text{ext}}) 3^{B_n}, \quad \mathcal{L}(B_n \mid \mathcal{F}_n) = \text{Bin}(s_n, 1/4), \quad [\text{S4.5}]$$

and therefore

$$v_3(\det K(D_n)) = v_3(\det K_n^{\text{ext}}) + B_n. \quad [\text{S4.6}]$$

On $\mathcal{G}_n$, the factorization and determinant identities hold with the shadow and exterior signs fixed for every internal sign vector. The binomial law applies when the internal vector is sampled. The slot positions and K_n^{ext} may be dependent, while the conditional law of the internal signs remains product-uniform.

S4.10. The maximal binomial atom. After conditioning on the shadow and the exterior signs, a prescribed determinant can force at most one value of B_n . Uniform determinant anti-concentration therefore reduces to a bound on the largest point mass of this binomial variable. The following elementary estimate is the final probabilistic input.

Lemma S4.10 (Binomial maximal atom). *If $B \sim \text{Bin}(m, 1/4)$, then*

$$\sup_{k \in \mathbb{Z}} \mathbb{P}\{B = k\} \leq \sqrt{\frac{2\pi}{3m}}.$$

Proof. The characteristic function is

$$\phi(t) = \frac{3}{4} + \frac{1}{4}e^{it},$$

and

$$|\phi(t)|^2 = 1 - \frac{3}{4} \sin^2(t/2).$$

Fourier inversion and $1 - x \leq e^{-x}$ give

$$\mathbb{P}\{B = k\} \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp\left(-\frac{3m}{8} \sin^2(t/2)\right) dt.$$

For $|t| \leq \pi$, $\sin(|t|/2) \geq |t|/\pi$. Extending the resulting Gaussian integral to $\mathbb{R}$ yields

$$\mathbb{P}\{B = k\} \leq \frac{1}{2\pi} \int_{\mathbb{R}} \exp\left(-\frac{3m}{8\pi^2} t^2\right) dt = \sqrt{\frac{2\pi}{3m}}.$$

$\square$

Theorem S4.11 (Trefoil-slot determinant anti-concentration). *There are constants $\delta, c_0, C_0 > 0$ and $n_0 \geq 4$ such that, for $n \geq n_0$ and a uniform rooted diagram D_n in either the rooted general or the rooted reduced Chapman class,*

$$\sup_{a \geq 1} \mathbb{P}\{\det K(D_n) = a\} \leq C_0 e^{-c_0 n} + \sqrt{\frac{2\pi}{3[\delta n]}} = O(n^{-1/2}).$$

The bound is uniform over moving determinant targets $a = a_n$.

Proof. Fix $a \geq 1$. On $\mathcal{G}_n$, condition on $\mathcal{F}_n$. If $\det K(D_n) = a$, Eq. (S4.6) forces

$$B_n = v_3(a) - v_3(\det K_n^{\text{ext}}).$$

Conditional on $\mathcal{F}_n$, the right-hand side is fixed, so at most one value of B_n is possible. Lemma S4.10 gives

$$\mathbf{1}_{\mathcal{G}_n} \mathbb{P}\{\det K(D_n) = a \mid \mathcal{F}_n\} \leq \mathbf{1}_{\mathcal{G}_n} \sqrt{\frac{2\pi}{3s_n}}.$$

Taking expectations and adding $\mathbb{P}(\mathcal{G}_n^c)$ from Lemma S4.3 proves the displayed bound, uniformly in a . $\square$

Corollary S4.12 (Determinant collision entropy). *For an independent copy D'_n ,*

$$\mathbb{P}\{\det K(D_n) = \det K(D'_n)\} = O(n^{-1/2}),$$

and

$$H_2(\det K(D_n)) \geq \frac{1}{2} \log n - O(1).$$

Proof. Writing $p_a = \mathbb{P}\{\det K(D_n) = a\}$,

$$\mathbb{P}\{\det K(D_n) = \det K(D'_n)\} = \sum_a p_a^2 \leq \sup_a p_a.$$

Apply Theorem S4.11 and take negative logarithms. $\square$

Theorem S4.11 is a moving-target statement. The stronger exponential estimate for a fixed determinant uses a different binomial lower-tail argument and is proved in Section S5.

S5. Fixed targets, determinant refinements, and bit complexity

The determinant anti-concentration theorem yields fixed-target, determinant-refinement, independent-database, and balanced-benchmark consequences. Determinant evaluation also has polynomial bit complexity.

S5.1. Fallback identities.

Proposition S5.1 (Black-hole and fallback identities). *Let (Ω_n, μ_n) be any discrete object ensemble and let I be a sound invariant. Fix any complete decision procedure **Decide** for the underlying equivalence problem and define the invariant-first algorithm*

$$A_I(x, y) = \begin{cases} \text{No}, & I(x) \neq I(y), \\ \text{Decide}(x, y), & I(x) = I(y). \end{cases}$$

The black hole of this algorithm is

$$\mathcal{B}_n(I) = \{(x, y) \in \Omega_n^2 : I(x) = I(y)\}.$$

If X_n, X'_n are independent with law μ_n , then

$$\mathbb{P}\{(X_n, X'_n) \in \mathcal{B}_n(I)\} = \sum_v \mathbb{P}\{I(X_n) = v\}^2 = \alpha_n(I). \quad [\text{S5.1}]$$

For a fixed target x_* , the corresponding one-object fallback set is

$$\mathcal{B}_n(I; x_*) = \{x \in \Omega_n : I(x) = I(x_*)\},$$

and its mass is the single atom

$$p_n(I; x_*) := \mathbb{P}\{X_n \in \mathcal{B}_n(I; x_*)\} = \mathbb{P}\{I(X_n) = I(x_*)\}. \quad [\text{S5.2}]$$

The random-pair probability averages over a second independent sample, whereas the fixed-target probability is the mass of one prescribed fiber.

For an observed database $x_1, \dots, x_M$, write $N_v = \#\{i : I(x_i) = v\}$. The all-pairs fallback count is

$$F_M(I) = \sum_v \binom{N_v}{2}. \quad [\text{S5.3}]$$

Pointwise, $F_M(I) = C_M(I)$. Under iid sampling, $\mathbb{E}F_M(I) = \binom{M}{2} \alpha_n(I)$. These identities count fallback calls only; they do not weight calls by the runtime of the complete decider.

Corollary S5.2 (Vanishing determinant black hole for a random pair). *For independent uniform rooted diagrams D_n, D'_n in either diagram model,*

$$\mathbb{P}\{(D_n, D'_n) \in \mathcal{B}_n(\det)\} = \alpha_n(\det) = O(n^{-1/2}).$$

Thus the determinant-first algorithm returns a sound negative answer before fallback with probability $1 - O(n^{-1/2})$.

Proof. Apply Proposition S5.1 and Corollary S4.12. $\square$

S5.2. Exponential rarity of a fixed determinant. Recall from Section S4 that $\mathcal{G}_n \in \mathcal{F}_n$ and, almost surely on the good-shadow event $\mathcal{G}_n$,

$$\det K(D_n) = \det(K_n^{\text{ext}})3^{B_n}, \quad \mathcal{L}(B_n \mid \mathcal{F}_n) = \text{Bin}(s_n, 1/4), \quad s_n = \lfloor \delta n \rfloor, \quad [\text{S5.4}]$$

where K_n^{ext} is $\mathcal{F}_n$ -measurable and $\mathbb{P}(\mathcal{G}_n^c) \leq C_0 e^{-c_0 n}$.

Theorem S5.3 (Fixed determinant values are exponentially rare). *For every fixed positive integer a , there are constants $C_a, c_a > 0$ such that, in either diagram model,*

$$\mathbb{P}\{\det K(D_n) = a\} \leq C_a e^{-c_a n}. \quad [\text{S5.5}]$$

Proof. Let $r = v_3(a)$. On $\mathcal{G}_n$, Eq. (S5.4) and nonnegativity of $v_3(\det K_n^{\text{ext}})$ imply

$$\det K(D_n) = a \implies B_n \leq r.$$

Since $\mathcal{G}_n \in \mathcal{F}_n$ and K_n^{ext} is $\mathcal{F}_n$ -measurable,

$$\mathbb{P}\{\det K(D_n) = a, \mathcal{G}_n\} \leq \mathbb{E}[\mathbf{1}_{\mathcal{G}_n} \mathbb{P}\{B_n \leq r \mid \mathcal{F}_n\}].$$

For $B \sim \text{Bin}(m, 1/4)$,

$$\begin{aligned} \mathbb{P}\{B \leq r\} &= \left(\frac{3}{4}\right)^m \sum_{j=0}^r \binom{m}{j} 3^{-j} \\ &\leq \left(\frac{3}{4}\right)^m \sum_{j=0}^r \frac{m^j}{3^j j!} \leq C_r m^r \left(\frac{3}{4}\right)^m. \end{aligned} \quad [\text{S5.6}]$$

By the conditional law in Eq. (S5.4), the conditional probability inside the expectation is exactly this binomial lower tail with $m = s_n$ on $\mathcal{G}_n$. Because r is fixed and $s_n \geq (\delta/2)n$ for all sufficiently large n , the polynomial factor can be absorbed into the exponential. For example, after increasing the multiplicative constant,

$$s_n^r \left(\frac{3}{4}\right)^{s_n} \leq C'_r \exp\left[-\frac{\delta}{4} \log\left(\frac{4}{3}\right) n\right].$$

Adding the exceptional probability $C_0 e^{-c_0 n}$ proves Eq. (S5.5) with

$$c_a = \min\left\{c_0, \frac{\delta}{4} \log\left(\frac{4}{3}\right)\right\}$$

and a target-dependent constant C_a . $\square$

Corollary S5.4 (Fixed targets and fixed knot types). *For every fixed knot K_* ,*

$$\mathbb{P}\{\det K(D_n) = \det K_*\} = O_{K_*}(e^{-c_{K_*} n}),$$

and hence

$$\mathbb{P}\{K(D_n) \cong K_*\} = O_{K_*}(e^{-c_{K_*} n}).$$

The determinant-first algorithm comparing D_n with K_ therefore enters its fallback with exponentially small probability.*

The fixed-target estimate follows from $B_n \leq v_3(\det K_*)$. For a random target, the corresponding valuation can lie in the central range of the binomial law, so the two-sample collision estimate is different.

S5.3. Determinant-refining invariants. The normalized Alexander polynomial determines the determinant through $\det K = |\Delta_K(-1)|$. Consequently, all diagrams with one prescribed Alexander polynomial lie in a single determinant fiber. The same argument applies to any invariant from which the determinant can be recovered, and the following definition records precisely that property.

Definition S5.5. A normalized invariant J is *determinant-refining* on a class of knots if there is a deterministic recovery map g such that

$$\det K = g(J(K))$$

for every knot in that class.

Proposition S5.6 (Monotonicity under deterministic refinement). *If J is determinant-refining, then*

$$p_n^{\max}(J) \leq p_n^{\max}(\det), \quad \alpha_n(J) \leq \alpha_n(\det). \quad [\text{S5.7}]$$

For every fixed normalized value w of J ,

$$\mathbb{P}\{J(D_n) = w\} \leq \mathbb{P}\{\det K(D_n) = g(w)\}. \quad [\text{S5.8}]$$

Consequently, every such J inherits the $O(n^{-1/2})$ random-pair collision estimate and exponential fixed-value rarity.

Proof. Every J -fiber is contained in one determinant fiber. The maximal-atom inequality follows immediately. If $q_w = \mathbb{P}\{J(D_n) = w\}$, then grouping the q_w over the determinant fibers gives

$$\sum_w q_w^2 \leq \sum_a \left(\sum_{g(w)=a} q_w \right)^2 = \alpha_n(\det),$$

which proves the collision inequality. Eq. (S5.8) is the same fiber inclusion for one value. $\square$

Table S4 lists determinant-refining invariants and the required normalization data. Grading-forgotten homology ranks or unspecified polynomial normalizations need not determine the determinant.

Table S4. Determinant-refining invariants and required normalizations. Refinement transfers the probability bounds but does not imply polynomial-time evaluation.

Fingerprint	Recovery of $\det K$	Normalization information that must be retained	Consequence
Determinant	Identity	Positive integer $ \Delta_K(-1) $	Polynomial-time evaluation
Normalized Alexander polynomial	$ \Delta_K(-1) $	Symmetric Laurent polynomial with $\Delta_K(1) = 1$ (5)	Collision and fixed-value bounds
Jones polynomial	$ V_K(-1) $	Standard oriented normalization with $V_{\bigcirc} = 1$ (6)	Collision and fixed-value bounds
HOMFLY-PT polynomial	Standard Alexander specialization, then $ \Delta_K(-1) $	Full normalized two-variable polynomial with $P_{\bigcirc} = 1$ and its skein-variable convention retained (7)	Collision and fixed-value bounds
Unreduced bigraded Khovanov homology	Divide the graded Euler characteristic $\chi_q(K)$ by $q + q^{-1}$, recover $V_K(t)$ with $t = q^2$, and take $ V_K(-1) $	Both gradings and the oriented normalization shift retained; $V_{\bigcirc} = 1$ (8)	Collision and fixed-value bounds
Bigraded knot Floer homology $\widehat{HFK}$	Graded Euler characteristic gives the symmetrized Alexander polynomial	Maslov and Alexander gradings retained (9)	Collision and fixed-value bounds

For the unreduced convention in Table S4, $\chi_q(K) = (q + q^{-1})V_K(q^2)$; form the Laurent-polynomial quotient by $q + q^{-1}$ before evaluation.

Grading-forgotten specializations, total ranks, a single coefficient, and isolated higher-color evaluations are absent from the table because they need not determine the determinant. A joint fingerprint containing a determinant-refining component is itself determinant-refining.

S5.4. Polynomial bit complexity of the determinant branch. We use a standard combinatorial input model: an n -crossing knot diagram is given by a planar-map or signed Gauss-type encoding of length polynomial in n . Checkerboard shading produces a reduced Goeritz matrix G_D whose absolute determinant is the knot determinant (10). The number of white regions is at most $n + 2$, so G_D has order at most $n + 1$. Every entry is a signed sum of crossing incidences and has absolute value at most n .

Proposition S5.7 (Determinant evaluation is polynomial-time). *The knot determinant of an n -crossing diagram can be computed in polynomial bit complexity.*

Proof. Constructing the checkerboard graph and G_D requires polynomial time. Apply fraction-free Gaussian elimination to G_D (11), with row or column pivoting when needed. The algorithm uses $O(n^3)$ exact integer arithmetic operations. Its intermediate entries are minors of the input matrix. If H is any $r \times r$ minor, Hadamard's inequality gives

$$|\det H| \leq \prod_{i=1}^r \|H_i\|_2 \leq (\sqrt{r}n)^r.$$

Since $r \leq n + 1$, every intermediate integer has $O(n \log n)$ bits. Addition, multiplication, exact division, and comparison of integers of this length all have polynomial bit cost, and there are polynomially many such operations. The final absolute determinant is therefore obtained in polynomial bit complexity. $\square$

The first branch has polynomial bit complexity. Complete knot equivalence is decidable by normal-surface classification procedures (12, 13), so the invariant-first procedure is a total algorithm. Results on the complexity of unknotting, splitting, genus, and knottedness concern related decision problems (14, 15). No runtime bound for the complete branch enters the present result. The estimate

$$\mathbb{P}\{\text{fallback}\} = O(n^{-1/2})$$

controls how often that branch is called, not its contribution to expected runtime; a sufficiently expensive rare branch can dominate an expectation. The algorithmic conclusion is that a sound negative answer is produced at polynomial cost on a set of random pairs of probability $1 - O(n^{-1/2})$.

S5.5. Independent databases.

Corollary S5.8 (Collision-free independent databases). *Let $D_n^{(1)}, \dots, D_n^{(M_n)}$ be independent uniform rooted diagrams from the complexity- n law in either diagram model, and let $F_{M_n}(\det)$ be the number of determinant-first all-pairs fallback calls. Then*

$$\mathbb{E}F_{M_n}(\det) = \binom{M_n}{2} \alpha_n(\det) = O(M_n^2 n^{-1/2}).$$

Consequently,

$$M_n = o(n^{1/4}) \implies \mathbb{P}\{F_{M_n}(\det) > 0\} \longrightarrow 0.$$

Proof. Use Proposition S5.1, Corollary S4.12, and Markov's inequality. $\square$

This is a sufficient range. The determinant theorem gives no lower bound on $\alpha_n(\det)$, so it gives no converse at $M \asymp n^{1/4}$.

S5.6. Balanced classification benchmarks. A classifier can achieve high accuracy on random knot pairs by using only a cheap incomplete invariant. It accepts pairs with equal invariant values and rejects all pairs with unequal values. Soundness gives perfect acceptance on equivalent pairs, while a small collision probability makes the rule correct on most inequivalent pairs. A balanced benchmark gives equal weight to these positive and negative classes, and the following proposition quantifies this invariant-only baseline.

Proposition S5.9 (Balanced accuracy of invariant-only classifiers). *Let D_n, D'_n be independent with law μ_n , let*

$$\mathcal{K}_n = \{K(D_n) \cong K(D'_n)\}, \quad q_n = \mathbb{P}(\mathcal{K}_n) < 1,$$

and let I be any sound invariant. Consider a balanced binary benchmark in which the positive class consists of equivalent pairs and the negative class has the law of (D_n, D'_n) conditioned on $\mathcal{K}_n^c$. The classifier

$$h_I(D, D') = \mathbf{1}_{\{I(D)=I(D')\}}$$

has balanced accuracy

$$\text{BAcc}(h_I) = 1 - \frac{1}{2} \frac{\alpha_n(I) - q_n}{1 - q_n} \geq 1 - \frac{\alpha_n(I)}{2}. \quad [\text{S5.9}]$$

Consequently, if $\alpha_n(I) \rightarrow 0$, then $\text{BAcc}(h_I) \rightarrow 1$. In the two rooted Chapman ensembles, the determinant and every normalized determinant-refining invariant satisfy

$$\text{BAcc}(h_I) = 1 - O(n^{-1/2}).$$

For a fixed target knot K_ , set*

$$p_n(I; K_*) = \mathbb{P}\{I(D_n) = I(K_*)\}, \quad r_n(K_*) = \mathbb{P}\{K(D_n) \cong K_*\} < 1.$$

On a balanced target-recognition benchmark whose positive class represents K_* and whose negative class is D_n conditioned on $K(D_n) \not\cong K_*$, the rule $g_{I,K_*}(D) = \mathbf{1}_{\{I(D)=I(K_*)\}}$ has

$$\text{BAcc}(g_{I,K_*}) = 1 - \frac{1}{2} \frac{p_n(I; K_*) - r_n(K_*)}{1 - r_n(K_*)} \geq 1 - \frac{p_n(I; K_*)}{2}. \quad [\text{S5.10}]$$

Hence every fixed determinant-refining target has balanced accuracy $1 - O(e^{-cn})$ under the corresponding invariant-only rule.

Proof. Equivalent knots have equal invariant values, so the true-positive rate of h_I is one. On the conditioned negative class, the false-positive probability is

$$\mathbb{P}\{I(D_n) = I(D'_n) \mid \mathcal{K}_n^c\} = \frac{\mathbb{P}\{I(D_n) = I(D'_n)\} - \mathbb{P}(\mathcal{K}_n)}{1 - q_n} = \frac{\alpha_n(I) - q_n}{1 - q_n}.$$

Averaging the true-positive and true-negative rates gives the equality in Eq. (S5.9). Since $\mathcal{K}_n \subseteq \mathcal{B}_n(I)$, we have $0 \leq q_n \leq \alpha_n(I)$. The map $q \mapsto (\alpha_n(I) - q)/(1 - q)$ is decreasing on this interval, and therefore

$$\frac{\alpha_n(I) - q_n}{1 - q_n} \leq \alpha_n(I),$$

which proves the claimed bound in terms of collision mass alone. The fixed-target identity is the same calculation with the inclusions

$$\{K(D_n) \cong K_*\} \subseteq \{I(D_n) = I(K_*)\}.$$

Since $0 \leq r_n(K_*) \leq p_n(I; K_*)$, the same monotonicity argument gives $\frac{p_n(I; K_*) - r_n(K_*)}{1 - r_n(K_*)} \leq p_n(I; K_*)$. The asymptotic conclusions follow from Corollary S5.2, Proposition S5.6, and Theorem S5.3. $\square$

The rates above concern the displayed invariant-only rules under the stated probability laws; they do not describe a particular trained network. High balanced accuracy alone therefore does not show that a classifier distinguishes pairs that the chosen fingerprint fails to separate. To test that question, restrict the negative class to inequivalent pairs with equal fingerprint values. For any classifier f and normalized fingerprint J , whenever the conditioning event has positive probability, define the rejection rate on collision-conditioned negative pairs by

$$\Pi_n(f; J) = \mathbb{P}\{f(D_n, D'_n) = \text{No} \mid J(D_n) = J(D'_n), K(D_n) \not\cong K(D'_n)\}.$$

This rate should be reported together with the classifier's acceptance rate on equivalent pairs, since the constant-No rule has $\Pi_n(f; J) = 1$. If J_2 refines J_1 , then $\{J_2(D_n) = J_2(D'_n)\} \subseteq \{J_1(D_n) = J_1(D'_n)\}$. Thus $\Pi_n(f; J)$ is the rejection probability among inequivalent pairs in the corresponding invariant-equality event.

S6. Collisions in a finite census

This section treats uniform sampling without replacement from a fixed finite census. The rooted random-diagram law of the main theorem is a different probability measure.

S6.1. Fiber profiles and the replacement correction. Let the census contain T distinct entries, partitioned into R nonempty invariant fibers of sizes

$$m_1, \dots, m_R, \quad \sum_{v=1}^R m_v = T.$$

The with-replacement and known-distinct pair-collision masses are

$$\alpha^{\text{wr}} := \sum_v \left(\frac{m_v}{T}\right)^2, \quad \alpha^{\text{neq}} := \frac{\sum_v m_v(m_v - 1)}{T(T - 1)}.$$

They satisfy the identity

$$\alpha^{\text{neq}} = \frac{T\alpha^{\text{wr}} - 1}{T - 1}, \quad \alpha^{\text{wr}} - \alpha^{\text{neq}} = \frac{1 - \alpha^{\text{wr}}}{T - 1}. \quad [\text{S6.1}]$$

Indeed, $T^2\alpha^{\text{wr}} = \sum_v m_v^2$ and $T(T - 1)\alpha^{\text{neq}} = \sum_v m_v^2 - T$.

S6.2. Mean and variance without replacement. Choose a uniform M -subset of the T entries and let C_M^{fp} be the number of unordered pairs in the subset that lie in a common fiber. Write

$$(x)_k = x(x-1)\cdots(x-k+1),$$

with $(x)_0 = 1$, and set

$$S_2 = \sum_v (m_v)_2, \quad S_3 = \sum_v (m_v)_3,$$

$$a = \frac{S_2}{(T)_2} = \alpha^{\text{neq}}, \quad b = \frac{S_3}{(T)_3}, \quad d = \frac{S_2^2 - 4S_3 - 2S_2}{(T)_4}.$$

Here b is the probability that two sampled pairs sharing one endpoint both collide, and d is the corresponding probability for two disjoint sampled pairs.

Theorem S6.1 (Finite-population collision moments). *For $T \geq 4$ and $0 \leq M \leq T$,*

$$\mathbb{E}C_M^{\text{fp}} = \binom{M}{2}a, \tag{S6.2}$$

$$\text{Var}(C_M^{\text{fp}}) = \binom{M}{2}(a - a^2) + 6\binom{M}{3}(b - a^2) + 6\binom{M}{4}(d - a^2). \tag{S6.3}$$

Terms whose binomial coefficient vanishes are omitted automatically.

Proof. Expose the sampled subset in a uniformly random order $Y_1, \dots, Y_M$ and put

$$I_{ij} = \mathbf{1}_{\{I(Y_i)=I(Y_j)\}}.$$

Every pair has collision probability a , proving Eq. (S6.2). Two indicators sharing one index are simultaneously one if and only if three distinct sampled entries lie in the same fiber, which has probability b .

For two disjoint pairs, the numerator of the joint-collision probability is

$$\sum_v (m_v)_4 + \sum_{v \neq w} (m_v)_2(m_w)_2.$$

The identity

$$(m)_2^2 - (m)_4 = 4(m)_3 + 2(m)_2$$

shows that this numerator equals $S_2^2 - 4S_3 - 2S_2$, giving d . The complete graph on M sampled positions contains $3\binom{M}{3}$ unordered pairs of edges sharing a vertex and $3\binom{M}{4}$ unordered pairs of disjoint edges. Covariances are doubled in the variance expansion, yielding Eq. (S6.3). $\square$

Corollary S6.2 (Relative variance). *For $T \geq 4$, $M \geq 2$, and $a > 0$,*

$$\frac{\text{Var}(C_M^{\text{fp}})}{(\mathbb{E}C_M^{\text{fp}})^2} \leq \frac{4}{M^2a} + \frac{8}{M\sqrt{a}} + \frac{20}{T}. \tag{S6.4}$$

Proof. The first variance contribution uses $a - a^2 \leq a$. Termwise, $(m)_3 \leq (m)_2^{3/2}$, and therefore

$$S_3 \leq S_2^{3/2}.$$

For $T \geq 4$ this gives

$$b \leq a^{3/2} \frac{(T)_2^{3/2}}{(T)_3} = a^{3/2} \frac{\sqrt{T(T-1)}}{T-2} \leq 2a^{3/2}.$$

Also,

$$d \leq \frac{S_2^2}{(T)_4} = a^2 \frac{T(T-1)}{(T-2)(T-3)},$$

so

$$d - a^2 \leq a^2 \frac{4T-6}{(T-2)(T-3)} \leq \frac{20}{T}a^2.$$

Dividing Eq. (S6.3) by $\binom{M}{2}^2 a^2$ and using

$$\binom{M}{2}^{-1} \leq \frac{4}{M^2}, \quad \frac{6\binom{M}{3}}{\binom{M}{2}^2} \leq \frac{4}{M}, \quad \frac{6\binom{M}{4}}{\binom{M}{2}^2} \leq 1$$

proves the claim. $\square$

The mean controls the regime in which collisions disappear, while the relative-variance bound controls the regime in which they occur with high probability. Together they give the finite-population birthday dichotomy.

Theorem S6.3 (Finite-population inverse birthday law). *Consider a sequence of finite censuses with $T_n \rightarrow \infty$, and choose a uniform M_n -subset from the n th census.*

(i) *If $M_n^2 \alpha_n^{\text{neq}} \rightarrow 0$, then*

$$\mathbb{P}\{C_{M_n}^{\text{fp}} > 0\} \rightarrow 0.$$

(ii) *If $M_n^2 \alpha_n^{\text{neq}} \rightarrow \infty$, then*

$$\mathbb{P}\{C_{M_n}^{\text{fp}} > 0\} \rightarrow 1.$$

Proof. The first statement follows from Markov's inequality and Eq. (S6.2). Under the second hypothesis, both $M_n^2 a_n$ and $M_n \sqrt{a_n}$ diverge, while $T_n \rightarrow \infty$. The right-hand side of Eq. (S6.4) therefore tends to zero. Chebyshev's inequality gives $\mathbb{P}\{C_{M_n}^{\text{fp}} = 0\} \rightarrow 0$. $\square$

S6.3. No-collision probabilities. A collision-free sample contains at most one entry from each invariant fiber. The number of such samples is encoded by an elementary symmetric polynomial. Let $e_M(m_1, \dots, m_R)$ denote the M th elementary symmetric polynomial, with $e_0 = 1$.

Theorem S6.4 (Finite-census no-collision probabilities). *For a uniform M -subset of the census,*

$$\mathbb{P}\{C_M^{\text{fp}} = 0\} = \frac{e_M(m_1, \dots, m_R)}{\binom{T}{M}}. \quad [\text{S6.5}]$$

For M independent draws with replacement from the same census,

$$\mathbb{P}\{C_M = 0\} = \frac{M! e_M(m_1, \dots, m_R)}{T^M}. \quad [\text{S6.6}]$$

Both probabilities are zero when $M > R$.

Proof. A collision-free subset chooses one entry from each of M distinct fibers. For a fixed set of fibers, the number of choices is the product of their sizes; summing over all fiber sets gives e_M . In ordered sampling with replacement, the same selected entries can appear in $M!$ orders. $\square$

The coefficients $e_0, \dots, e_R$ can be computed in $O(TR)$ integer operations by multiplying

$$\prod_{v=1}^R (1 + m_v z)$$

coefficient by coefficient. Eq. (S6.5) therefore determines the probability for every M . A percentile such as

$$M_{50} = \min\{M : \mathbb{P}(C_M^{\text{fp}} > 0) \geq 1/2\}$$

is then obtained by the integer comparison $2e_M \leq \binom{T}{M}$.

S6.4. The extremal profile at fixed support. Among profiles with the same T and R , the balanced profile maximizes the no-collision probability at every database size.

Theorem S6.5 (Balanced-fiber extremum). *Fix integers $1 \leq R \leq T$ and let $m = (m_1, \dots, m_R)$ range over positive integer vectors with sum T . Write*

$$T = qR + r, \quad 0 \leq r < R,$$

and let

$$b = (\underbrace{q+1, \dots, q+1}_r, \underbrace{q, \dots, q}_{R-r})$$

be the balanced profile. Then:

(i) *b minimizes α^{neq} , with minimum*

$$\alpha_{\min}^{\text{neq}}(T, R) = \frac{Rq(q-1) + 2rq}{T(T-1)}; \quad [\text{S6.7}]$$

(ii) *for every $0 \leq M \leq T$,*

$$e_M(m_1, \dots, m_R) \leq e_M(b_1, \dots, b_R), \quad [\text{S6.8}]$$

so the balanced profile maximizes the no-collision probability for every database size M .

2689	<i>Proof.</i> Suppose two coordinates satisfy $x \geq y + 2$. Replace them by $x - 1$ and $y + 1$. This preserves positivity and the total sum,	2753
2690	while	2754
2691	$x^2 + y^2 - [(x - 1)^2 + (y + 1)^2] = 2(x - y - 1) > 0.$	2755
2692	Hence repeated balancing decreases $\sum_v m_v^2$ until all coordinates differ by at most one. Since	2756
2693		2757
2694	$\alpha^{\text{neq}} = \frac{\sum_v m_v^2 - T}{T(T - 1)},$	2758
2695		2759
2696		2760
2697	this proves part (i), and direct substitution gives Eq. (S6.7).	2761
2698	For part (ii), the cases $M = 0$ and $M = 1$ are immediate. Assume $M \geq 2$, hold the remaining $R - 2$ coordinates fixed, and	2762
2699	write their elementary symmetric polynomials as E_j . Then	2763
2700	$e_M(x, y, \text{others}) = xyE_{M-2} + (x + y)E_{M-1} + E_M.$	2764
2701		2765
2702	The balancing move fixes $x + y$ and increases the product by	2766
2703		2767
2704	$(x - 1)(y + 1) - xy = x - y - 1 > 0.$	2768
2705		2769
2706	Since $E_{M-2} \geq 0$, the move cannot decrease e_M . Iteration reaches the balanced profile, proving Eq. (S6.8). □	2770
2707		2771
2708	Corollary S6.6 (Bound from support size). <i>For every fiber profile,</i>	2772
2709		2773
2710	$\alpha^{\text{neq}} \geq \frac{T - R}{R(T - 1)}.$	2774
2711		2775
2712	<i>Equality in this Cauchy bound requires equal fibers and therefore can occur only when R divides T.</i>	2776
2713		2777
2714	<i>Proof.</i> Cauchy's inequality gives $\sum_v m_v^2 \geq T^2/R$. Substitute into $\alpha^{\text{neq}} = (\sum_v m_v^2 - T)/[T(T - 1)]$. □	2778
2715		2779
2716	If a census sequence satisfies	2780
2717	$\frac{T_n - R_n}{T_n - 1} \geq \eta > 0,$	2781
2718	then	2782
2719	$\alpha_n^{\text{neq}} \geq \frac{\eta}{R_n},$	2783
2720		2784
2721	and databases with $M_n^2/R_n \rightarrow \infty$ collide with high probability. Balanced profiles with $T_n/R_n \rightarrow \rho > 1$ have $\alpha_n^{\text{neq}} = \Theta(R_n^{-1})$, so	2785
2722	their birthday scale has order $\sqrt{R_n}$. Near-injective sequences fall outside this conclusion. When $R = T$, every fiber is a singleton,	2786
2723	$\alpha^{\text{neq}} = 0$, and a database sampled without replacement never contains an invariant collision.	2787
2724		2788
2725	<i>Remark S6.7</i> (Sampling law). Throughout this section, the underlying population is a fixed census. The principal law is uniform	2789
2726	sampling without replacement; the displayed companion formula treats independent draws with replacement. Chapman's rooted	2790
2727	random-diagram measure and a distribution on knot types define different probability laws and require separate analyses.	2791
2728		2792
2729	S7. Census data and a finite benchmark	2793
2730	The examples below are hypothetical finite profiles, not knot-census data, and do not estimate the Chapman-model limit. We	2794
2731	first list the information required for an empirical census calculation.	2795
2732		2796
2733	S7.1. Data required for a census calculation. An empirical census calculation should record:	2797
2734		2798
2735	(i) the source census, version, retrieval date, and checksum;	2799
2736		2800
2737	(ii) whether entries are knots, links, prime types, mirror-identified types, or diagram representatives;	2801
2738		2802
2739	(iii) the crossing range and cumulative versus exact-crossing convention;	2803
2740		2804
2741	(iv) a stable object identifier and duplicate policy;	2805
2742		2806
2743	(v) the normalization and equality convention for every invariant;	2807
2744		2808
2745	(vi) the exact fiber table $\{(v, m_v)\}$;	2809
2746		2810
2747	(vii) exact-integer outputs for T , R , largest fiber, singleton counts, α^{wr} , α^{neq} , and e_M ;	2811
2748		2812
2749	(viii) a script regenerating every table and vector figure.	2813
2750		2814
2751		2815
2752		2816

Table S5. Two finite fiber profiles. M_{50} is the smallest database size whose collision probability is at least one half.

Profile	R	R/T	α^{wr}	α^{neq}	M_{50}
$A = (101, 1^{899})$	900	0.90	111/10000	101/9990	17
$B = (2^{500})$	500	0.50	1/500	1/999	38

Support size and collision probability

Both profiles contain $T = 1000$ entries. Profile A has larger support, but $M_{50}(A) = 17 < 38 = M_{50}(B)$.

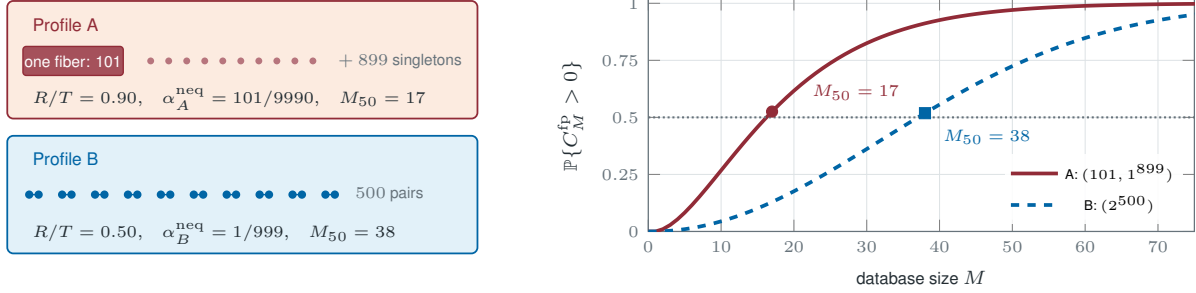


Fig. S3. Two fiber profiles in a finite population. For uniform sampling without replacement from $T = 1000$ entries, the collision probability is $1 - e_M(m_1, \dots, m_R)/\binom{T}{M}$. Profile A has fibers $(101, 1^{899})$ and reaches probability one half at $M_{50} = 17$. Profile B has fibers (2^{500}) and reaches it at $M_{50} = 38$. Although profile A has more occupied values, its larger collision mass gives $M_{50}(A) < M_{50}(B)$. These are hypothetical profiles, not knot-census data.

S7.2. A finite profile. Larger occupied support need not imply a larger median collision threshold: one heavy fiber can decrease M_{50} .

Proposition S7.1 (Support and database capacity). *Consider the two fiber profiles on $T = 1000$ entries*

$$A = (101, 1^{899}), \quad B = (2^{500}).$$

Profile A occupies more invariant values than profile B, but

$$\alpha_A^{\text{neq}} = \frac{101}{9990} > \frac{1}{999} = \alpha_B^{\text{neq}}, \quad M_{50}(A) = 17 < 38 = M_{50}(B).$$

Thus occupied support and database capacity can rank fingerprints in opposite orders.

The first profile has one large fiber, whereas every fiber of the second has size two. Table S5 gives their values.

Proof. The elementary-symmetric numerators have closed forms:

$$e_M(A) = \binom{899}{M} + 101 \binom{899}{M-1},$$

$$e_M(B) = 2^M \binom{500}{M}.$$

Consequently,

$$\mathbb{P}_A\{C_M^{\text{fp}} = 0\} = \frac{\binom{899}{M} + 101 \binom{899}{M-1}}{\binom{1000}{M}},$$

$$\mathbb{P}_B\{C_M^{\text{fp}} = 0\} = \frac{2^M \binom{500}{M}}{\binom{1000}{M}}.$$

Exact integer comparison gives

$$\mathbb{P}_A\{C_{16}^{\text{fp}} = 0\} > \frac{1}{2}, \quad \mathbb{P}_A\{C_{17}^{\text{fp}} = 0\} < \frac{1}{2},$$

and

$$\mathbb{P}_B\{C_{37}^{\text{fp}} = 0\} > \frac{1}{2}, \quad \mathbb{P}_B\{C_{38}^{\text{fp}} = 0\} < \frac{1}{2}.$$

These inequalities prove the proposition. $\square$

S7.3. Programs and data. The script

```
python code/generate_finite_benchmarks.py -out-dir data
```

computes elementary symmetric polynomials by exact integer dynamic programming, evaluates all probabilities as rational numbers, and exports

- data/finite_profile_summary.csv, and
- data/finite_no_collision_curves.csv.

The test program enumerates small finite populations and checks the expectation, variance, no-collision formula, balanced-fiber extremum, support bound, and benchmark percentiles.

S8. Adaptive invariant cascades

An adaptive cascade applies a first invariant to every sampled entry and reserves later invariants for entries that still share a joint fingerprint. An entry that is alone in its current sampled bucket leaves the cascade; pairs that remain together after the last stage are sent to the terminal procedure.

The population is a fixed set Ω of T distinct entries, and the database $S \subset \Omega$ is a uniform M -subset. Every expectation is therefore with respect to sampling without replacement from one finite population. The invariant values themselves are deterministic labels on Ω .

S8.1. Sample buckets, singleton objects, and unresolved pairs. Let J be a fingerprint with population fibers $F_{J,v}$ of sizes $m_{J,v}$. The sampled J -bucket of value v is $S \cap F_{J,v}$. For a sampled database S , write

$$N_{J,v}(S) = |S \cap F_{J,v}|.$$

Define the three sample-level statistics

$$S_J^{\text{obs}}(S) = \sum_v N_{J,v}(S) \mathbf{1}_{\{N_{J,v}(S)=1\}}, \quad [\text{S8.1}]$$

$$U_J^{\text{obs}}(S) = \sum_v N_{J,v}(S) \mathbf{1}_{\{N_{J,v}(S) \geq 2\}} = M - S_J^{\text{obs}}(S), \quad [\text{S8.2}]$$

$$C_J(S) = \sum_v \binom{N_{J,v}(S)}{2}. \quad [\text{S8.3}]$$

Thus S_J^{obs} counts entries already unique inside the sampled database, U_J^{obs} counts entries that must proceed to a refinement, and C_J counts unresolved pairs.

Lemma S8.1 (Expected singleton and survivor counts). For $1 \leq M \leq T$,

$$S_J(M) := \mathbb{E}S_J^{\text{obs}} = \frac{1}{\binom{T}{M}} \sum_v m_{J,v} \binom{T - m_{J,v}}{M - 1}, \quad [\text{S8.4}]$$

$$U_J(M) := \mathbb{E}U_J^{\text{obs}} = M - S_J(M). \quad [\text{S8.5}]$$

Proof. Fix an entry $x \in F_{J,v}$. A sampled M -subset contains x and no other entry of its J -fiber in exactly $\binom{T - m_{J,v}}{M - 1}$ ways: after selecting x , all remaining entries must be chosen from the $T - m_{J,v}$ entries outside $F_{J,v}$. Dividing by $\binom{T}{M}$ and summing the indicator over the $m_{J,v}$ entries of every fiber gives Eq. (S8.4). Eq. (S8.5) follows pointwise from Eq. (S8.2). $\square$

Lemma S8.2 (Expected unresolved-pair count). For $0 \leq M \leq T$,

$$P_J(M) := \mathbb{E}C_J = \frac{\binom{M}{2}}{\binom{T}{2}} \sum_v \binom{m_{J,v}}{2} = \binom{M}{2} \alpha_J^{\text{neq}}. \quad [\text{S8.6}]$$

Proof. Every unordered pair of distinct population entries belongs to a uniform M -subset with probability $\binom{M}{2} / \binom{T}{2}$. Summing over the $\sum_v \binom{m_{J,v}}{2}$ equal- J population pairs proves the formula. $\square$

Both lemmas depend only on the fibers of the joint fingerprint J and remain valid for arbitrarily dependent component invariants.

S8.2. Expected workload. Let $I_1, \dots, I_k$ be invariants and

$$J_j = (I_1, \dots, I_j)$$

their prefix fingerprints. The adaptive policy evaluates I_1 on every sampled entry. For $j \geq 2$, it evaluates I_j exactly on the entries lying in nonsingleton sampled J_{j-1} -buckets. Since J_j refines J_{j-1} , pairs separated by one prefix remain separated by every later prefix, so no skipped evaluation can affect the final unresolved-pair set.

Assume a deterministic cost $c_j \geq 0$ per evaluation of I_j and a deterministic cost $f \geq 0$ for each pair passed to the terminal complete procedure. For a realized sample S , the workload is

$$W(S) = Mc_1 + \sum_{j=2}^k c_j U_{J_{j-1}}^{\text{obs}}(S) + fC_{J_k}(S). \quad [\text{S8.7}]$$

Theorem S8.3 (Expected cascade workload). *Under the finite-population law above,*

$$\mathbb{E}W = Mc_1 + \sum_{j=2}^k c_j U_{J_{j-1}}(M) + fP_{J_k}(M). \quad [\text{S8.8}]$$

Proof. Take expectations in Eq. (S8.7) and apply Lemmas S8.1 and S8.2. $\square$

Eq. (S8.8) assumes constant costs. For object-dependent evaluation costs, the object indicators may instead be weighted by their individual costs. If the terminal cost is nonlinear in bucket size, fC_{J_k} must be replaced by the corresponding bucket functional. The main text uses the constant-cost formula above.

S8.3. Ordering and marginal break-even rules. Consider two invariants A and B . Compare policies that evaluate both coordinates adaptively and then invoke the same terminal procedure on the same joint fingerprint (A, B) . Their expected workloads are

$$\begin{aligned} \mathbb{E}W_{A \rightarrow B} &= Mc_A + c_B U_A(M) + fP_{A,B}(M), \\ \mathbb{E}W_{B \rightarrow A} &= Mc_B + c_A U_B(M) + fP_{A,B}(M). \end{aligned}$$

Proposition S8.4 (Two-stage ordering criterion). *The order $A \rightarrow B$ is no more expensive than $B \rightarrow A$ exactly when*

$$c_A S_B(M) \leq c_B S_A(M). \quad [\text{S8.9}]$$

When both expected singleton counts are positive, this is equivalent to

$$\frac{c_A}{S_A(M)} \leq \frac{c_B}{S_B(M)}. \quad [\text{S8.10}]$$

Proof. Using $U_A = M - S_A$ and $U_B = M - S_B$,

$$\mathbb{E}W_{A \rightarrow B} - \mathbb{E}W_{B \rightarrow A} = c_A S_B(M) - c_B S_A(M).$$

The common joint-fallback term cancels. $\square$

Eq. (S8.10) ranks two-stage orders by cost per expected sample singleton resolved. For longer cascades, minimization uses the full objective in Eq. (S8.8); pairwise greedy ordering need not minimize it.

Now fix a prefix J and consider adding one more invariant I before the same terminal pairwise fallback. Without I , the terminal expected cost is $fP_J(M)$. With I , the additional evaluation cost is $c_I U_J(M)$ and the terminal expected cost is $fP_{J,I}(M)$.

Proposition S8.5 (Marginal break-even rule). *Adding I after J does not increase expected cost exactly when*

$$c_I U_J(M) \leq f[P_J(M) - P_{J,I}(M)]. \quad [\text{S8.11}]$$

If $P_J(M) > P_{J,I}(M)$, the critical fallback cost is

$$f_{\text{crit}}(I \mid J) = \frac{c_I U_J(M)}{P_J(M) - P_{J,I}(M)}. \quad [\text{S8.12}]$$

If the new coordinate removes no unresolved pairs, it cannot reduce workload at positive evaluation cost.

Proof. Subtract the expected workload without I from the workload with I and rearrange. Since (J, I) refines J , the pair reduction in Eq. (S8.11) is nonnegative. $\square$

S8.4. A nested finite example. The nested profile below gives both the number of entries evaluated and the number of pairs reaching the terminal procedure. It also gives numerical instances of the ordering and break-even criteria. The population contains $T = 1000$ labeled entries. At the first stage, 899 entries are singleton fibers and the remaining 101 entries form one heavy fiber. The second coordinate divides 100 entries of that heavy fiber into twenty quintuplets and leaves the last entry singleton. The third coordinate divides each quintuplet into one pair and three singletons. Thus the prefix profiles are

$$J_1 = (101, 1^{899}), \quad J_2 = (5^{20}, 1^{900}), \quad J_3 = (2^{20}, 1^{960}). \quad [\text{S8.13}]$$

In the explicit object labels, I_2 determines I_1 and I_3 determines I_2 . Hence an I_2 -only policy has the J_2 fibers, and an $I_1 \rightarrow I_3$ policy has the J_3 fibers even though the intermediate coordinate is not evaluated. The corresponding numbers of equal-prefix population pairs are

$$Q_{J_1} = 5050, \quad Q_{J_2} = 200, \quad Q_{J_3} = 20. \quad [\text{S8.14}]$$

For $M = 100$, Table S6 gives the expected counts. The singleton numerators follow directly from Eq. (S8.4); for example,

$$S_{J_1}(100) = \frac{899 \binom{999}{99} + 101 \binom{999}{99}}{\binom{1000}{100}},$$

$$S_{J_2}(100) = \frac{900 \binom{999}{99} + 100 \binom{995}{99}}{\binom{1000}{100}},$$

$$S_{J_3}(100) = \frac{960 \binom{999}{99} + 40 \binom{998}{99}}{\binom{1000}{100}}.$$

Table S6. Sample-level expectations in the finite cascade. Decimals are shown only for readability; the associated CSV files retain the rational numerators and denominators.

Prefix	Occupied values	Population pairs	$U_J(100)$	$P_J(100)$
J_1	900	5050	10.099835	$5555/111 = 50.045045$
J_2	920	200	3.417052	$220/111 = 1.981982$
J_3	980	20	$44/111 = 0.396396$	$22/111 = 0.198198$

To compare policies, assign the illustrative constant costs

$$(c_1, c_2, c_3) = (1, 12, 40), \quad f = 100.$$

These values are part of the synthetic example and are not empirical timing measurements. Substitution in Eq. (S8.8) gives Table S7.

Table S7. Expected workload for selected policies in the finite example.

Policy	$\mathbb{E}W$
I_1 then fallback	5104.5045
I_2 then fallback	1398.1982
$I_2 \rightarrow I_1$ then fallback	1401.6152
$I_1 \rightarrow I_2$ then fallback	419.3962
$I_1 \rightarrow I_3$ then fallback	523.8132
$I_1 \rightarrow I_2 \rightarrow I_3$ then fallback	377.6999
all three coordinates on all entries	5319.8198

For the first two coordinates,

$$c_1 S_{J_2}(100) = 96.5829 < 1078.8020 = c_2 S_{J_1}(100),$$

so Proposition S8.4 shows that $I_1 \rightarrow I_2$ has lower expected cost. The break-even thresholds are

$$f_{\text{crit}}(I_2 \mid J_1) = 2.521646,$$

$$f_{\text{crit}}(I_3 \mid J_2) = 76.624792.$$

At $f = 100$, both break-even inequalities hold; the reduction obtained from the third stage is smaller.

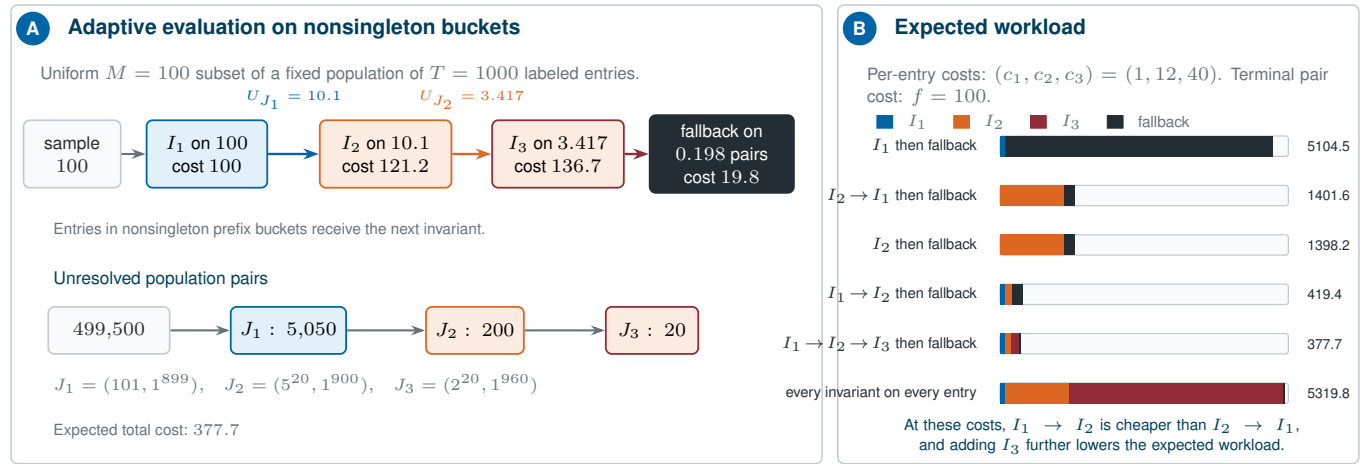


Fig. S4. Expected workload for an adaptive cascade on an explicitly labeled population of $T = 1000$ entries. The first invariant is evaluated on every sampled entry; each later invariant is evaluated only on entries lying in nonsingleton buckets of the preceding prefix fingerprint. The joint values available after stage j form $J_j = (I_1, \dots, I_j)$. (A) The numbers of population pairs unresolved by J_1, J_2, J_3 are 5050, 200, and 20, respectively. In a uniform $M = 100$ subset, the expected numbers of entries evaluated by I_2 and I_3 are 10.1 and 3.417, respectively, and the expected number of pairs reaching the terminal procedure is 0.198. (B) The bars give invariant-evaluation and terminal costs for $(c_1, c_2, c_3, f) = (1, 12, 40, 100)$. The adaptive cascade has expected cost 377.7, compared with 5104.5 for I_1 followed by the terminal procedure and 5319.8 when every invariant is evaluated on every entry. This hypothetical population is not knot-census data.

S8.5. Programs and data. The command

```
python code/generate_cascade_benchmarks.py -out-dir data
```

regenerates:

- `cascade_population.csv`, containing all 1000 object-level labels;
- `cascade_summary.csv`, containing exact singleton, survivor, and pair expectations;
- `cascade_workloads.csv`, containing exact policy costs; and
- `cascade_ordering_break_even.csv`, containing the ordering and marginal thresholds.

The test program enumerates small sampled populations and checks Eqs. (S8.4), (S8.6), (S8.8), (S8.9), and (S8.11). All displayed benchmark values use integers and rational arithmetic.

The example is a hypothetical finite population. An application to a knot census would supply the object identifiers, invariant normalizations, duplicate policy, and generating program specified in Section S7.

S9. Computations and source files

The figures, supporting programs, and combined manuscript and Supporting Information PDF are available in the project repository at github.com/pedro9olivares/knots-blackholes-databases-birthdays.

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